Multi-Target Tracking for Satellite Videos Guided by Spatial-Temporal Proximity and Topological Relationships

Jianzhi Hong, Taoyang Wang, Yuqi Han, and Tong Wei

***Abstract*— The features of moving targets in satellite videos are sparse and similar, resulting in two major challenges for multi-target tracking: detection losses and association errors. The rich spatial-temporal proximity and topological relationships among targets in satellite videos can indicate the feature enhancement of small targets and generate discriminative individual descriptions of targets, which helps to improve the accuracy of multi-target tracking. In this article, we propose a novel Multi-target Tracking method for satellite videos guided by Spatial-Temporal Proximity and Topological Relationships (MTT-STPTR). Specifically, a spatial-temporal relationship sparse attention (STRSA) module is constructed in the feature extraction stage to accurately enhance the feature expression of small targets by capturing the cross-frame semantics relevant to the target areas. In addition, a joint feature matching (JFM) module is designed in the interframe association stage, which constructs a novel similarity measurement method of star-shaped topological structures and uses it to measure the similarity of multidimensional features of target individuals, thereby alleviating association errors caused by dense individuals with similar features. Moreover, a novel multiple granularity spatial-temporal contrastive learning (MGSTCL) module is designed to promote a balanced optimization of detection and association tasks for multicategory targets in satellite videos. Experiments conducted on two public datasets, VISO and AIR-MOT, demonstrate that MTT-STPTR outperforms existing state-of-the-art multi-target tracking methods in terms of the multiobject-tracking accuracy (MOTA) and identification $F1$-score (IDF1), indicating its effectiveness.**

***Index Terms*— Contrastive learning, multi-target tracking, optical satellite video, spatial-temporal proximity, topological relationship.**

I. Introduction

COMPARED with drone-based videos and ground surveillance videos, high-resolution optical satellite videos (HROSV) offer the advantage of continuously monitoring larger areas in a staring mode [1], [2], [3]. Multi-target tracking based on HROSV can extract the movement trajectories of a large number of ground targets over a period of time and plays an important role in fields such as traffic perception, military reconnaissance, and commercial analysis. Therefore, it has attracted the attention of many researchers [4], [5], [6].

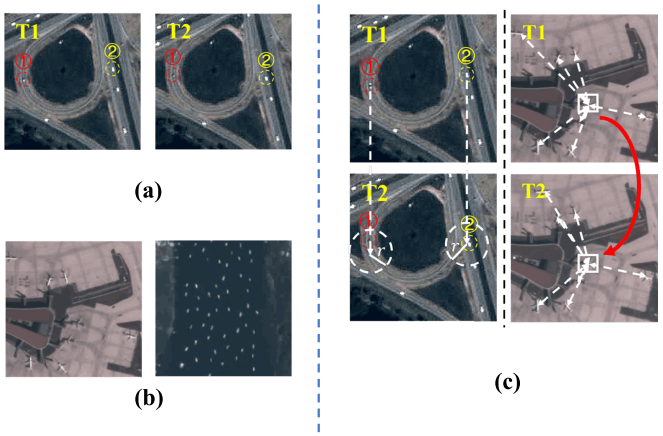


Fig. 1. Challenges of MTT on satellite videos and the motivation of the proposed work. (a) Low contrast between target and background (①) and small size (②). (b) Sparse textures and dense distribution. (c) Motivation of our work: (left) spatial-temporal proximity and (right) topological relationships between targets.

Multi-target tracking typically comprises two subtasks: target detection and interframe association [7]. Based on the coupling of these subtasks, existing multi-target tracking methods can be categorized into two main paradigms: tracking-by-detection (TBD) and joint detection and tracking (JDT). Compared with TBD [8], [9], [10], [11], [12], [13], [14], [15], JDT reduces the dependence on high-precision detection results and offers greater computational efficiency. Therefore, JDT has attracted the attention of researchers. For further details, please refer to Section II. However, the JDT methods designed for general videos are not fully suitable for satellite videos. There are two significant challenges in multi-target tracking for satellite videos.

1) The sparse features of small targets pose detection difficulties [see Fig. 1(a)]. Satellite videos are captured far away from the Earth, and the moving targets (such as cars, boats, and airplanes) have low spatial resolution, small size, sparse texture features, low contrast with

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Jianzhi Hong, Taoyang Wang, and Tong Wei are with the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan 430079, China (e-mail: hongjianzhi@whu.edu.cn; wangtaoyang@whu.edu.cn; weitong@whu.edu.cn).

Yuqi Han is with the School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China, and also with the National Key Laboratory of Science and Technology on Space-Born Intelligent Information Processing, Beijing 100081, China (e-mail: yuqi_han@bit.edu.cn).

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the background, motion blur, and so on. These factors make it difficult for targets to form the stable and discriminative appearance feature expressions.

2) The similarity of individual features leads to association errors [see Fig. 1(b)]. Interframe association requires that the feature representation of the same individual remains consistent and unique over time. The long-distance bird's-eye view used in satellite videos contributes to the loss of texture and perspective features, leading to confusion among individuals within local areas.

In terms of small target detection, JDT currently mainly adopts two strategies: multiscale feature extraction [16], [17] and historical information compensation. However, targets in satellite videos are extremely small (the size of the vehicle is 5×5 pixels), so the continuous downsampling in the multiscale feature extraction method still leads to information loss and missed detection, indicating that it is impossible to form effective feature expression by relying solely on single-frame information. To address this, some methods focus on information compensation. Studies [18], [19], [20], [21], [22], [23] have introduced multiframe features to enhance feature expression. However, these approaches primarily involve splicing and fusing multiframe feature maps, which limits performance improvements. In contrast, other methods [7], [24], [25], [26], [27] use the attention mechanism to align feature pixels from adjacent frames, allowing for targeted compensation of historical information. Nonetheless, these methods mainly search for the target area for information compensation on a global scale, leading to issues of erroneous compensation and increased computational complexity, ultimately constraining detection improvements.

In addition, *in terms of interframe association,* appearance feature matching and motion feature matching are two primary solutions. For satellite-video targets with scarce texture features, it is inefficient and unreliable to use the feature vectors of the detection area as clues for interframe association [16], [20] because they cannot effectively distinguish individual differences. In this regard, motion features of targets (such as displacement [7], [18], [19], [28] and speed [29], [30], [31]) offer some individual distinguishability, but they are easy to fail in local scenes with dense targets. In addition, some studies have achieved robust interframe association by measuring the similarity of relationships between targets, known as topological features. These existing topological similarity measurement methods are mainly categorized into two types: target similarity [32], [33], [34], [35], [36], [37], [38] and position similarity [39], [40]. However, the former methods lack the use of spatial position relationships within frames, limiting their effectiveness for satellite-video targets with sparse texture features. The latter methods depend on the assumption that the target distribution in adjacent frames remains stable. In contrast to ground and airborne videos, satellite videos frequently contain numerous small moving targets, and the appearance and disappearance of these targets can lead to instability in target distribution. Therefore, when modeling features, it is crucial to consider not only the one-to-one adjacency relationships between targets but also the similarity of the overall topological structure to enhance stability amid changes in target distribution.

In summary, accurate compensation of historical information and target association in dense scenes are two key challenges for applying the JDT method to high-precision multi-target tracking in satellite videos. *As for Challenge 1*, considering the high frame rate of satellite videos (generally above 10 frames/s) and the slight displacement of targets between the feature maps of adjacent frames, we believe that there are potential targets in the current frame located in the neighborhood of targets from the previous frame, as shown in the left part of Fig. 1(c). Therefore, by restricting the target area requiring information compensation in the current frame to the neighborhood of the target in the previous frame, we can achieve precise feature enhancement while significantly reducing computational costs. *As for Challenge 2*, we have observed that the bird's-eye view in satellite videos allows for clear extraction of positional relationships and star-shaped topological structures among targets. These relationships are unique to each target and remain relatively stable over short periods. Since the star-shaped topological structure feature is formed around each individual target, its feature can be easily compensated to the target's individual features, thereby enhancing the distinguishability of individual features and achieving accurate association of dense targets, as shown in the right part of Fig. 1(c). In addition, the competitive learning between detection and association presents a common issue for JDT methods, that is, *Challenge 3*. Satellite videos contain various types of moving targets, such as cars, boats, airplanes, and trains. The detection task emphasizes expanding interclass variance, while the optimization goal of the association task focuses on increasing intraclass variance. Consequently, when intraclass variance exceeds interclass variance, detection performance may decline and vice versa. Therefore, a new optimization goal must be established to balance the relationship between interclass and intraclass variance.

Based on the above analysis, this article proposes a novel Multi-target Tracking method for satellite videos guided by Spatial-Temporal Proximity and Topological Relationships (MTT-STPTR). MTT-STPTR follows the JDT paradigm and aims to utilize the relationship between targets to achieve feature enhancement and accurate interframe association, ultimately enabling high-precision extraction of target trajectories in satellite videos. MTT-STPTR comprises three innovative key modules: the spatial-temporal relationship sparse attention (STRSA) module, the multiple granularity spatial-temporal contrastive learning (MGSTCL) module, and the joint feature matching (JFM) module. For STRSA, we design a historical-trajectory-guided sparse attention module in the feature extraction stage to efficiently enhance features in the current frame. This module aggregates historical trajectory information and enhances the target features of the current frame by constructing a bipartite graph of the target region features from adjacent frames for multilayer attention calculation. For MGSTCL, we design a novel multigranularity contrast loss function to construct positive and negative sample pairs of adjacent frames based on targets from both individual and category granularities. This approach simultaneously improves

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the accuracy of target detection and interframe association. As for JFM, we developed a new method for measuring similarity in target individual features, compensated by the novel features of the star-shaped topological structure. This method is weighted and fused with appearance, motion, and topological similarity to obtain more complete and discriminative clues for individual association, thereby achieving accurate target association between frames. Through the comprehensive utilization of these three modules, our proposed method MTT-STPTR can achieve robust tracking of multi-targets in satellite videos.

The main contributions of this article are summarized as follows.

1) We propose a novel multi-target tracking method for satellite videos called MTT-STPTR. Unlike other JDT methods, MTT-STPTR incorporates spatial-temporal proximity and topological relationships between targets to enhance feature propagation and individual feature modeling, enabling stable tracking of small and dense multi-targets.
2) STRSA is proposed in the feature extraction stage to accurately capture the cross-frame semantics related to target areas based on the spatial-temporal proximity of targets, effectively enhancing the feature representation of small targets.
3) JFM is proposed in the interframe association stage to alleviate association errors caused by the similarity of individual features, which measures the similarity of the star-shaped topological structures of targets in adjacent frames and constructs a comprehensive joint feature description that includes appearance, motion, and topology.
4) A novel multigranularity contrastive learning loss function is proposed, which improves the distinguishability of target features at both individual and category granularities. This approach promotes a balanced optimization between detection and association tasks, addressing the challenges posed by multicategory targets in satellite videos.

The remainder of this article is organized as follows. Section II introduces the related work of multi-target tracking based on JDT. Section III introduces our proposed method MTT-STPTR in detail. Section IV presents the experimental data and evaluation metrics. Section V presents and analyses the experimental results of MTT-STPTR. Finally, we conclude the article in Section VI.

II. Related Work

Multi-target tracking based on JDT adopts the one-stage multitask parallel processing approach, which can detect targets of interest and simultaneously extract individual features of targets or realize data association. We review the research on JDT on general video and satellite video in Sections II-A and II-B, respectively.

A. Multi-Target Tracking for General Videos Based on JDT

In terms of multi-target tracking for general videos, CenterTrack [18] introduces an offset prediction branch based on the anchor-free detector CenterNet, which extracts the relative offsets of targets between previous and subsequent frames, facilitating interframe association through motion feature matching. PermaTrack [19] also incorporates gated recurrent units into CenterTrack to filter and aggregate historical feature information, enhancing its antiocclusion capabilities. FairMOT [16] adds a reidentification branch to the output head to extract the appearance features of individuals, thereby realizing interframe association based on appearance feature matching. Some studies [7] have identified a conflict in optimization goals between the detection and reidentification subtasks, which can undermine multi-target tracking accuracy. To address this issue, TraDeS proposes to use motion cues to improve the performance of detection and achieve interframe association, thereby alleviating the optimization conflict between these subtasks.

B. Multi-Target Tracking for Satellite Videos Based on JDT

The utilization of time-series information in the methods described above has provided significant inspiration for the advancement of multi-target tracking in satellite videos. To address the optimization conflict between detection and association, TGraM [20] introduces a spatial-temporal graph reasoning method along with a multitask learning balance strategy to enhance the accuracy of multi-target tracking. CFTracker [24] improves feature representation and tracking continuity of small targets in satellite videos through methods such as time-series feature enhancement and cross-frame consistency training. In addition, Zhang et al. [14] proposed a knowledge distillation method suitable for multi-target tracking to improve the efficiency of model training. Zhao et al. [25] present a JDT framework named MP2Net, which incorporates a multiframe multiscale attention module to improve the accuracy of small target extraction. MGTrack [26] achieves stable tracking of high-speed moving targets by generating offsets and guiding historical features for alignment and fusion. MCTrack [27] generates more discriminative individual feature vectors through multiscale spatial-temporal feature fusion, thereby improving the detection and tracking performance of small targets in satellite videos.

In summary, existing JDT methods for satellite-video processing mainly use the idea of historical information compensation to improve the detection performance of small targets. They utilize the cross-attention mechanism to perform pixel-by-pixel matching and fusion of multiple video frames, enabling targeted compensation of historical information and improvement of target features. However, these methods generally search for target areas for information compensation across a global scope. In satellite videos with complex backgrounds and wide fields of view, this approach can lead to issues such as error compensation and complex computations, ultimately resulting in limited improvements in detection performance. Moreover, existing methods primarily identify individuals and achieve interframe association based on the similarity of appearance and motion. However, in satellite videos where targets exhibit sparse features and dense distributions, these methods are prone to association errors and losses.

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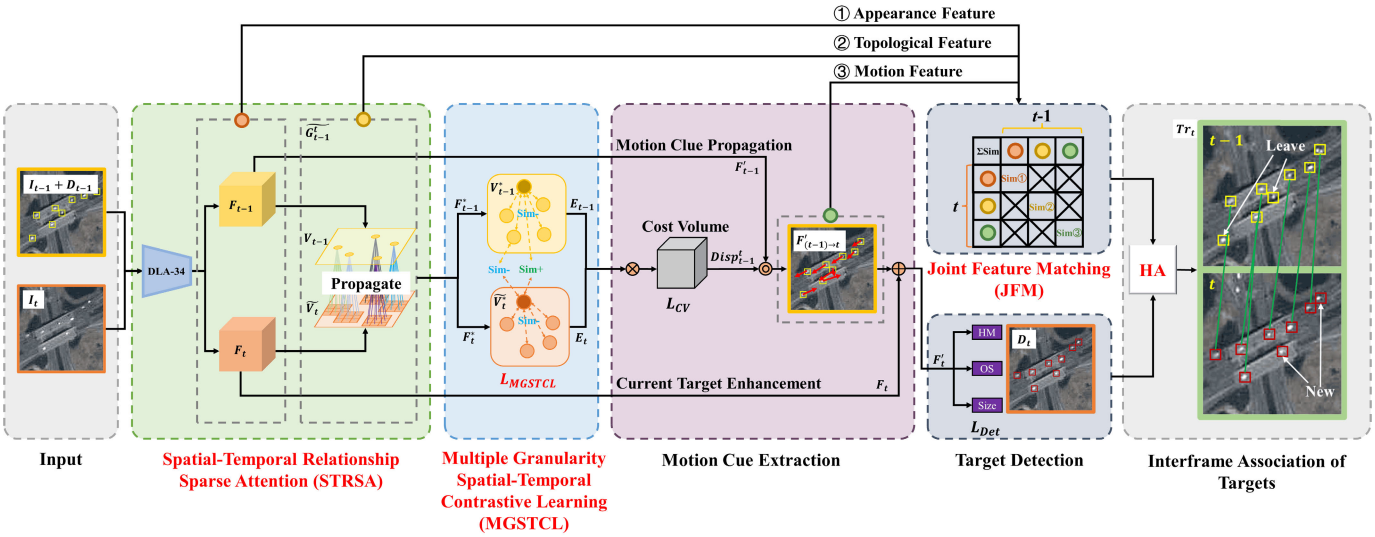


Fig. 2. Framework of the MTT-STPTR algorithm.

Based on the above analysis, this work aims to guide the precise compensation of historical information by learning the spatial-temporal proximity relationships between targets, thereby enhancing the feature representation of small targets. We design a multigranularity contrast loss function to alleviate the competition between category and individual features in JDT methods. In addition, we construct a topological relationship description between targets, which is utilized for the similarity measurement of multidimensional features, ultimately enabling the extraction of stable trajectories for moving cluster targets.

III. Methodology

In this section, we provide a detailed description of our proposed method, MTT-STPTR. The overall structure of the method is presented in Section III-A. Sections III-B–III-D introduce the details of our proposed STRSA, MGSTCL, and JFM, respectively.

A. Overview

As mentioned above, MTT-STPTR aims to introduce the spatial-temporal and topological relationships between targets for feature propagation and enhancement, as well as individual feature modeling, thereby improving the accuracy of multi-target tracking in satellite videos. Fig. 2 illustrates the framework of MTT-STPTR, including the STRSA module, the MGSTCL module, motion cue extraction, target detection, the JFM module, and interframe association of targets. The method takes two adjacent frames, $\boldsymbol{I}_{t-1}$ and $\boldsymbol{I}_t$, along with the detection results of the previous frame, $\boldsymbol{D}_{t-1}$, as inputs. It outputs the detection result $\boldsymbol{D}_t$ and the tracking result $\boldsymbol{Tr}_t$ of the current frame, thus completing the multi-target tracking process in satellite videos.

First, $\boldsymbol{I}_{t-1}$, $\boldsymbol{I}_t$, and $\boldsymbol{D}_{t-1}$ are input into the proposed STRSA to extract the appearance features $\boldsymbol{F}_{t-1}$ and $\boldsymbol{F}_t$, and the corresponding enhanced feature maps $\boldsymbol{F}^*_{t-1}$ and $\boldsymbol{F}^*_t$ are obtained through the interaction of spatial-temporal information. The sizes of $\boldsymbol{I}_{t-1}$ and $\boldsymbol{I}_t$ are represented as $3 \times \boldsymbol{h_I} \times \boldsymbol{w_I}$, while the sizes of $\boldsymbol{F}^*_{t-1}$ and $\boldsymbol{F}^*_t$ are $64 \times \boldsymbol{h_F} \times \boldsymbol{w_F}$. A detailed description of STRSA can be found in Section III-B.

Then, $\boldsymbol{F}^*_{t-1}$ and $\boldsymbol{F}^*_t$ are input into the proposed MGSTCL to enhance the distinguishability of target features at both individual and category granularities. This process yields the corresponding embedding vector sets $\boldsymbol{E}_{t-1}$ and $\boldsymbol{E}_t$ that are obtained, each with a size of $128 \times \boldsymbol{h_F} \times \boldsymbol{w_F}$. A detailed description of MGSTCL is provided in Section III-C. Following this, the cost volume $\boldsymbol{CV}^t_{t-1}$ is generated through the matrix multiplication of $\boldsymbol{E}_{t-1}$ and $\boldsymbol{E}_t$. The sizes of $\boldsymbol{CV}^t_{t-1}$ are $h_F \times w_F \times h_F \times w_F$. The calculation process can be expressed as follows:

$$\boldsymbol{E}_{t-1}, \boldsymbol{E}_t = \mathrm{MGSTCL}\left(\boldsymbol{F}^*_{t-1}, \boldsymbol{F}^*_t\right) \tag{1}$$

$$\boldsymbol{CV}^t_{t-1} = \boldsymbol{E}_{t-1} \otimes \boldsymbol{E}_t. \tag{2}$$

Referring to TraDeS [7], the pixel-by-pixel offset matrix of two adjacent frames $\boldsymbol{Disp}^t_{t-1}$ can be extracted based on $\boldsymbol{CV}^t_{t-1}$, and 3×3 deformable convolution kernel $\mathrm{DCN}_{3\times3}$ is constructed to transform the enhanced feature of targets $\boldsymbol{F}'_{t-1}$ into $\boldsymbol{F}'_{(t-1)\to t}$. $\boldsymbol{F}'_{t-1\to t}$ is then weightedly fused with $\boldsymbol{F}_t$ to obtain the appearance feature map enhanced by motion cues $\boldsymbol{F}'_t$, thus completing the motion cue extraction. Among them, $\boldsymbol{F}'_{t-1}$ is derived from the Hadamard product of the heatmap from the previous frame's detection results, $\boldsymbol{H}_{t-1}$, and the feature map $\boldsymbol{F}_{t-1}$. The size of $\boldsymbol{Disp}^t_{t-1}$ is $2 \times h_F \times w_F$, while the sizes of $\boldsymbol{F}'_{t-1}$, $\boldsymbol{F}'_{(t-1)\to t}$, and $\boldsymbol{F}'_t$ are all $64 \times h_F \times w_F$. $\boldsymbol{CV}^t_{t-1}$ is supervised by the loss function L_{CV} during training. The calculation process can be expressed as follows:

$$\boldsymbol{F}'_{t-1} = \boldsymbol{H}_{t-1} \circ \boldsymbol{F}_{t-1} \tag{3}$$

$$\boldsymbol{F}'_{(t-1)\to t} = \mathrm{DCN}_{3\times3}\left(\boldsymbol{Disp}^t_{t-1}, \boldsymbol{F}'_{t-1}\right) \tag{4}$$

$$\boldsymbol{F}'_t = w_{\boldsymbol{F}'_{(t-1)\to t}} * \boldsymbol{F}'_{(t-1)\to t} + w_{\boldsymbol{F}_t} * \boldsymbol{F}_t. \tag{5}$$

Among them, $w_{\boldsymbol{F}'_{(t-1)\to t}}$ and $w_{\boldsymbol{F}_t}$ represent the weights of feature fusion, while their sum is equal to 1.

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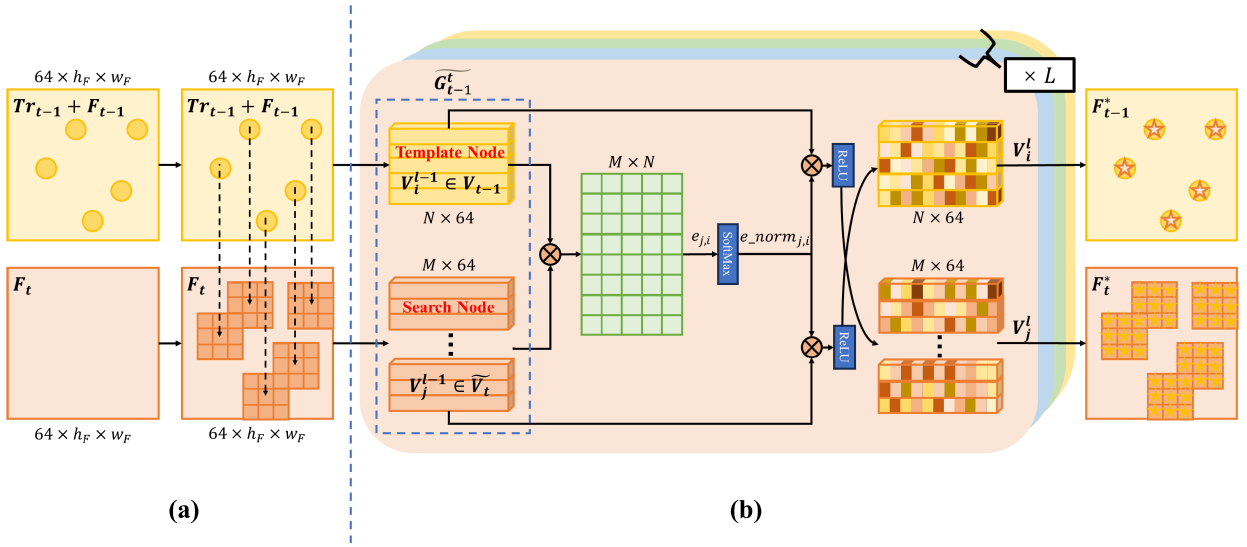


Fig. 3. STRSA. (a) Construction of interframe graph guided by historical trajectory. (b) Node feature enhancement based on graph attention aggregation.

Then, after $\boldsymbol{F}_t'$ is processed by the three output heads of heatmap, offset, and size, the detection result $\boldsymbol{D}_t$ of satellite videos is obtained, thus completing the target detection. During the training process, the three output heads will be supervised by the detection loss L_{DET} [41].

At the same time, the proposed JFM extracts the appearance and motion features of targets corresponding to the previous and the current frames, respectively. It also describes the topological features of neighboring targets to measure the similarity of target features between frames. A detailed description of JFM is provided in Section III-D.

Finally, based on the similarity matrix after weighted summation, the optimal allocation is performed through the Hungarian algorithm (HA) [42] to complete the interframe association between targets and obtain the multi-target tracking result of the current frame $\boldsymbol{Tr}_t$.

B. Spatial-Temporal Relationship Sparse Attention

The weak features of satellite-video targets make detection difficult. Existing JDT methods mainly solve this problem through historical information compensation strategies. However, they typically search for the target area for information compensation on a global scale, which can easily lead to errors in compensation and limited improvement in detection performance. Therefore, it is necessary to optimize the search strategy for target areas. STRSA is proposed to use historical trajectories to indicate the potential target area of the current frame and calculate the attention score of the interframe relationship between targets. This achieves a sparse representation of spatial-temporal features and accurate feature enhancement, thereby improving the accuracy of target extraction.

1) Interframe Graph Construction Guided by Historical Trajectory: We know the target position of the previous frame but not the target position of the current frame. Due to the low resolution and high frame rate of satellite videos, the feature map of the moving target after downsampling in adjacent frames will only show a small displacement. Therefore, it can be inferred that there are potential targets of the current frame in the neighborhood of targets of the previous frame (i.e., the historical trajectory). Based on this analysis, we construct an undirected bipartite graph $\widetilde{\boldsymbol{G}_{t-1}^t}$ to describe the interframe relationship between targets. Specifically, we take the feature vector set of the center positions of targets in the previous frame $\boldsymbol{V}_{t-1}$ and the neighborhood feature vector set corresponding to the current frame $\widetilde{\boldsymbol{V}}_t$ as two sets of nodes, as shown in Fig. 3(a). In addition, the nodes between groups are adjacent to each other, while the nodes within each group are not adjacent, resulting in the edge set $\boldsymbol{E}$. $\boldsymbol{E}$ and $\widetilde{\boldsymbol{G}_{t-1}^t}$ can be expressed as follows:

$$\boldsymbol{E} = \left\{ (\boldsymbol{V}_i, \boldsymbol{V}_j) \mid \forall \boldsymbol{V}_i \in \boldsymbol{V}_{t-1} \quad \forall \boldsymbol{V}_j \in \widetilde{\boldsymbol{V}}_t \boldsymbol{V}_{t-1} \cap \widetilde{\boldsymbol{V}}_t = \phi \right\} \tag{6}$$

$$\widetilde{\boldsymbol{G}_{t-1}^t} = \left(\boldsymbol{V}_{t-1}, \widetilde{\boldsymbol{V}}_t, \boldsymbol{E} \right). \tag{7}$$

2) Feature Enhancement of Targets Based on Graph Attention Aggregation: As shown in Fig. 3(b), $\boldsymbol{V}_{t-1}$ and $\widetilde{\boldsymbol{V}}_t$ can be regarded as template nodes and search nodes, respectively, after obtaining $\widetilde{\boldsymbol{G}_{t-1}^t}$. The target node to be enhanced is then determined by the attention scores between $\boldsymbol{V}_{t-1}$ and $\widetilde{\boldsymbol{V}}_t$. The higher the attention score, the greater the similarity between the search node and the template node, indicating a higher likelihood that they represent the same target. This is consistent with the idea of using the motion model to obtain a local search area and calculating the cross-correlation between the search area and the target template in single-target-tracking methods [43], [44], [45], [46], [47]. Based on this analysis, graph attention aggregation is introduced to guide the enhancement of target features. First, for each edge $(\boldsymbol{V}_j, \boldsymbol{V}_i) \in \boldsymbol{E}$, its attention value $e_{j,i}$ is expressed as

$$e_{j,i} = A(\boldsymbol{V}_j, \boldsymbol{V}_i) = (\boldsymbol{W}_j \boldsymbol{V}_j)(\boldsymbol{W}_i \boldsymbol{V}_i)^{\mathrm{T}}. \tag{8}$$

Among them, A is the graph attention module. $\boldsymbol{V}_j$ and $\boldsymbol{V}_i$ represent the search node and template node, respectively. $\boldsymbol{W}_j$ and $\boldsymbol{W}_i$ are the corresponding linear mapping matrices.

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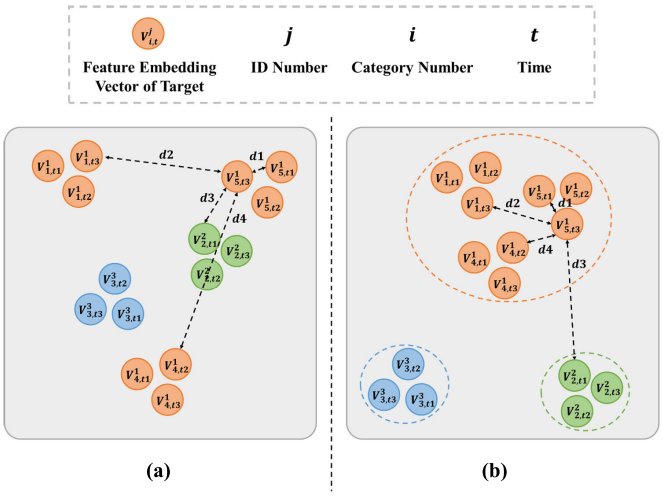


Fig. 4. Effect of contrastive learning at different granularities (targets of different colors represent different categories). (a) Contrastive learning at the individual granularity only. (b) Contrastive learning at the category-individual granularity (Ours).

$e_\mathrm{norm}_{j,i}$ is the normalized result of $e_{j,i}$

$$e_\mathrm{norm}_{j,i} = \mathrm{softmax}(e_{j,i}) = \frac{\exp(e_{j,i})}{\sum_{V_k \in V_{t-1}} \exp(e_{j,k})}. \quad (9)$$

$e_\mathrm{norm}_{j,i}$ represents the similarity between $\boldsymbol{V_j}$ and $\boldsymbol{V_i}$, which can be used to guide the node feature $\boldsymbol{V_i^{l-1}}$ at time $t-1$ to aggregate and update the node feature $\boldsymbol{V_j^l}$ at time t

$$\boldsymbol{V_j^l} = \mathrm{ReLU}\left(\sum_{V_i \in V_{t-1}} e_\mathrm{norm}_{j,i} \boldsymbol{W_h} \boldsymbol{V_i^{l-1}}\right), \quad l \in [1, L]. \quad (10)$$

Among them, l represents the number of the current calculation layer after the node feature propagation. L is the total number of graph attention layers, which is set to 2 in this case. In addition, for each edge $(\boldsymbol{V_i}, \boldsymbol{V_j}) \in \boldsymbol{E}$, its feature update process also refers to (8)–(10). After several calculations through the graph attention layers, the features of the targets in the current frame can be effectively enhanced.

C. Multiple Granularity Spatial-Temporal Contrastive Learning

There are several categories of moving targets in satellite videos, including cars, ships, airplanes, and trains. Since JDT simultaneously completes target detection and interframe association in a multitask learning manner, balancing the learning competition of multiple tasks is an ongoing challenge for JDT. Target detection aims to enhance the interclass differences in target features, while interframe association focuses on increasing the individual differences within target features. When the individual differences of targets within the same category exceed the interclass differences, it can lead to a decline in detection performance. As shown in Fig. 4(a), the distance $d4$ between different individuals of the same category is greater than the distance $d3$ between individual targets of different categories, resulting in interference between the features of different tasks.

Therefore, it is essential to design a new optimization goal. MGSTCL is proposed, which constructs a spatial-temporal contrast loss function based on both individual granularity and category granularity. This approach aims to minimize the distance between positive sample pairs while maximizing the distance between negative sample pairs, ensuring that the individual differences within the same category remain smaller than the differences between categories.

1) Individual-Granularity Spatial-Temporal Contrast Loss: For the feature of a target $\boldsymbol{V}$ in two adjacent frames, the features of targets with the same ID are considered positive samples $\boldsymbol{K_{ind}^+}$, while the features of targets with different IDs are regarded as negative samples $\boldsymbol{K_{ind}^-}$. The contrast loss function can then be designed as follows:

$$L_{\mathrm{ind}} = \ln\left[1 + \sum_{\boldsymbol{K_{ind}^+}} \sum_{\boldsymbol{K_{ind}^-}} \exp\left(\boldsymbol{V} \cdot \boldsymbol{K_{ind}^-} - \boldsymbol{V} \cdot \boldsymbol{K_{ind}^+}\right)\right]. \quad (11)$$

Among them, L_{ind} is always greater than or equal to 0. $\boldsymbol{V} \cdot \boldsymbol{K_{ind}^-}$ and $\boldsymbol{V} \cdot \boldsymbol{K_{ind}^+}$ represent the similarities of negative and positive sample pairs, respectively. When $\boldsymbol{V} \cdot \boldsymbol{K_{ind}^-}$ is smaller and $\boldsymbol{V} \cdot \boldsymbol{K_{ind}^+}$ is larger, that is, the features of negative sample pairs are more dissimilar and those of positive sample pairs are more similar, the individual loss L_{ind} is reduced, thereby enhancing the performance of interframe association.

2) Category-Granularity Spatial-Temporal Contrast Loss: Similar to the individual loss, for the feature of a target $\boldsymbol{V}$ in two adjacent frames, the features of targets belonging to the same category are considered positive samples $\boldsymbol{K_{cat}^+}$, while the features of targets from different categories are regarded as negative samples $\boldsymbol{K_{cat}^-}$. The category loss L_{cat} can be expressed by (12). The smaller L_{cat} is, the higher the classification accuracy of the target will be

$$L_{\mathrm{cat}} = \ln\left[1 + \sum_{\boldsymbol{K_{cat}^+}} \sum_{\boldsymbol{K_{cat}^-}} \exp\left(\boldsymbol{V} \cdot \boldsymbol{K_{cat}^-} - \boldsymbol{V} \cdot \boldsymbol{K_{cat}^+}\right)\right]. \quad (12)$$

The multigranularity spatial-temporal contrast loss L_{MGSTCL} is obtained through a weighted addition of the individual loss and the category loss, as shown in (13). During the training process, L_{MGSTCL} is used for back propagation. When L_{MGSTCL} approaches 0, the conditions $\boldsymbol{V} \cdot \boldsymbol{K_{ind}^+} > \boldsymbol{V} \cdot \boldsymbol{K_{ind}^-}$ and $\boldsymbol{V} \cdot \boldsymbol{K_{cat}^+} > \boldsymbol{V} \cdot \boldsymbol{K_{cat}^-}$ will be satisfied. As shown in Fig. 4(b), the distance $d1$ between the same individuals is smaller than the distances $d2$ and $d4$ between different individuals of the same category, while $d2$ and $d4$ are smaller than the distance $d3$ between individuals of different categories. In this way, the optimal balance between the detection-related features and the association-related features is achieved, thereby improving the accuracy of multi-target tracking

$$L_{\mathrm{MGSTCL}} = 0.5 * L_{\mathrm{ind}} + 0.5 * L_{\mathrm{cat}}. \quad (13)$$

D. Interframe Association Based on JFM

The similarity of individual features of targets in satellite videos can easily lead to association errors. Existing JDT methods primarily associate targets between frames based on appearance and motion feature matching, but these approaches

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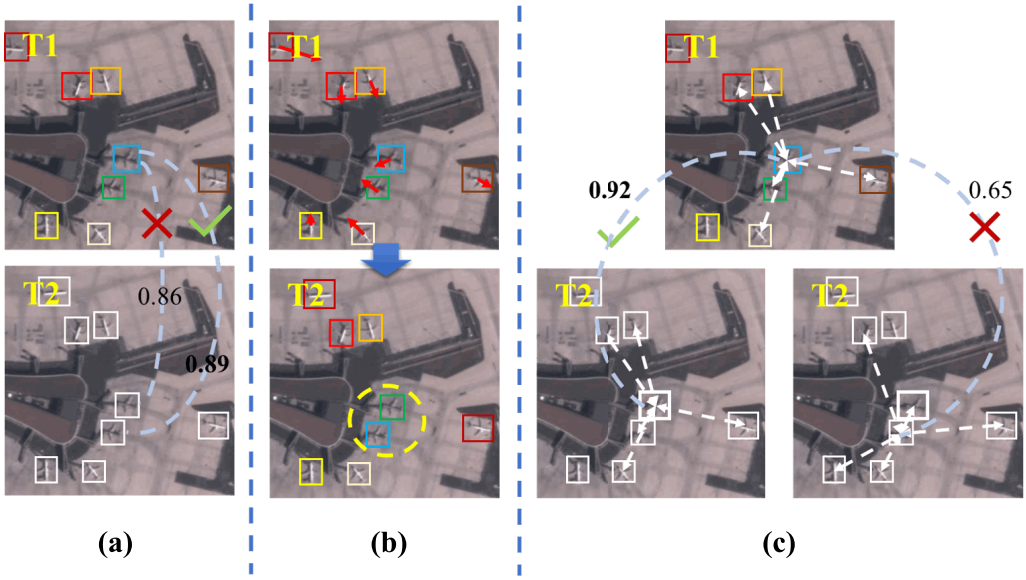


Fig. 5. Comparison of different association cues. (a) Appearance feature matching. (b) Motion feature matching. (c) Topological feature matching.

often result in target confusion in multi-target scenes characterized by sparse textures and dense distributions. As shown in Fig. 5(a), the appearance feature similarity of the same individual is less than that of different individuals. The identities of the two aircraft targets in Fig. 5(b) are also mistakenly swapped.

Therefore, it is necessary to explore other feature descriptions to enhance the distinguishability of individuals. We propose an interframe association method based on JFM that utilizes topological features to supplement individual feature descriptions. This approach enables robust multi-target association in satellite videos and improves tracking accuracy, as shown in Fig. 5(c).

1) Star-Shaped Topological Structure Description and Similarity Measurement: After obtaining the detection result of the current frame, we can identify the positions of all targets in both the previous and current frames, allowing us to describe the star-shaped topological structure of each target and its neighboring targets within the frame, which remains stable and unique over short periods. As shown in Fig. 6(a), we divide the neighboring targets into several subgraphs. For example, with the jth target $\boldsymbol{V}_{t,j}$ of the tth frame as the center, there are M targets closest to it (M is set to 5) divided into the same subgraph. As shown in Fig. 6(b), the edge features between the target $\boldsymbol{V}_{t,j}$ and its neighboring target $\boldsymbol{V}_{t,m}$ can be described as

$$\boldsymbol{T}_{t,j,m} = \left[\frac{x_m}{w_0}, \frac{y_m}{h_0}, \frac{x_m - x_j}{w_0}, \frac{y_m - y_j}{h_0}\right] \in R^4, \quad m \in [0, M]. \tag{14}$$

Among them, $\boldsymbol{T}_{t,j,m}$ represents the absolute position of the neighboring target $\boldsymbol{V}_{t,m}$ and its relative position to the target $\boldsymbol{V}_{t,j}$. x_j and y_j represent the horizontal and vertical coordinates of the target $\boldsymbol{V}_{t,j}$, respectively, while x_m and y_m represent the horizontal and vertical coordinates of the neighboring target $\boldsymbol{V}_{t,m}$. In addition, w_0 and h_0 indicate the width and height of the tth frame, respectively. Similarly, the edge feature representation $\boldsymbol{T}_{t-1,i,k}$ of the subgraph $\boldsymbol{G}_{t-1,i}$ centered on the target $\boldsymbol{V}_{t-1,i}$ can also be obtained.

As shown in (15), we use the squared Euclidean distance $\mathrm{ED}^2(\boldsymbol{T}_{t,j,m}, \boldsymbol{T}_{t-1,i,k})$ to evaluate the edge feature similarity between two subgraphs

$$\begin{aligned}\mathrm{ED}^2&\left(\boldsymbol{T}_{t,j,m}, \boldsymbol{T}_{t-1,i,k}\right) \\ &= \left(\frac{x_m}{w_0} - \frac{x_k}{w_0}\right)^2 + \left(\frac{y_m}{h_0} - \frac{y_k}{h_0}\right)^2 \\ &\quad + \left(\frac{x_m - x_j}{w_0} - \frac{x_k - x_i}{w_0}\right)^2 + \left(\frac{y_m - y_j}{h_0} - \frac{y_k - y_i}{h_0}\right)^2.\end{aligned} \tag{15}$$

When the value of $\mathrm{ED}^2(\boldsymbol{T}_{t,j,m}, \boldsymbol{T}_{t-1,i,k})$ is closer to 0, it indicates a higher probability that the neighboring target $\boldsymbol{V}_{t,m}$ belonging to the target $\boldsymbol{V}_{t,j}$ and the neighboring target $\boldsymbol{V}_{t-1,k}$ belonging to the target $\boldsymbol{V}_{t-1,i}$ are the same individual. This suggests that there are common neighboring targets in the subgraph $\boldsymbol{G}_{t,j}$ and the subgraph $\boldsymbol{G}_{t-1,i}$. When the number of common targets in these subgraphs increases, the similarity between them strengthens, resulting in a higher probability that $\boldsymbol{V}_{t,j}$ and $\boldsymbol{V}_{t-1,i}$ represent the same individual. Based on this analysis, we can describe the difference between the subgraphs as the normalized total cost $C_{\mathrm{norm}}(\boldsymbol{G}_{t,j}, \boldsymbol{G}_{t-1,i})$ after matching each element between subgraphs, as shown in (16) and (17) and Fig. 6(c). The similarity of subgraphs can be expressed as $\mathrm{EDSim}(\boldsymbol{T}_{t,j}, \boldsymbol{T}_{t-1,i})$, as shown in (18). The matching method employs greedy allocation, and the normalization technique utilizes the maximum and minimum normalization method, denoted as MaxMin$N$

$$C(\boldsymbol{G}_{t,j}, \boldsymbol{G}_{t-1,i}) = \sum_{(m',k') \in \boldsymbol{GM}} \mathrm{ED}^2\left(\boldsymbol{T}_{t,j,m}, \boldsymbol{T}_{t-1,i,k}\right) \tag{16}$$

$$C_{\mathrm{norm}}(\boldsymbol{G}_{t,j}, \boldsymbol{G}_{t-1,i}) = \mathrm{MaxMin}N\left(C\left(\boldsymbol{G}_{t,j}, \boldsymbol{G}_{t-1,i}\right)\right) \tag{17}$$

$$\mathrm{EDSim}\left(\boldsymbol{T}_{t,j}, \boldsymbol{T}_{t-1,i}\right) = 1 - C\left(\boldsymbol{G}_{t,j}, \boldsymbol{G}_{t-1,i}\right). \tag{18}$$

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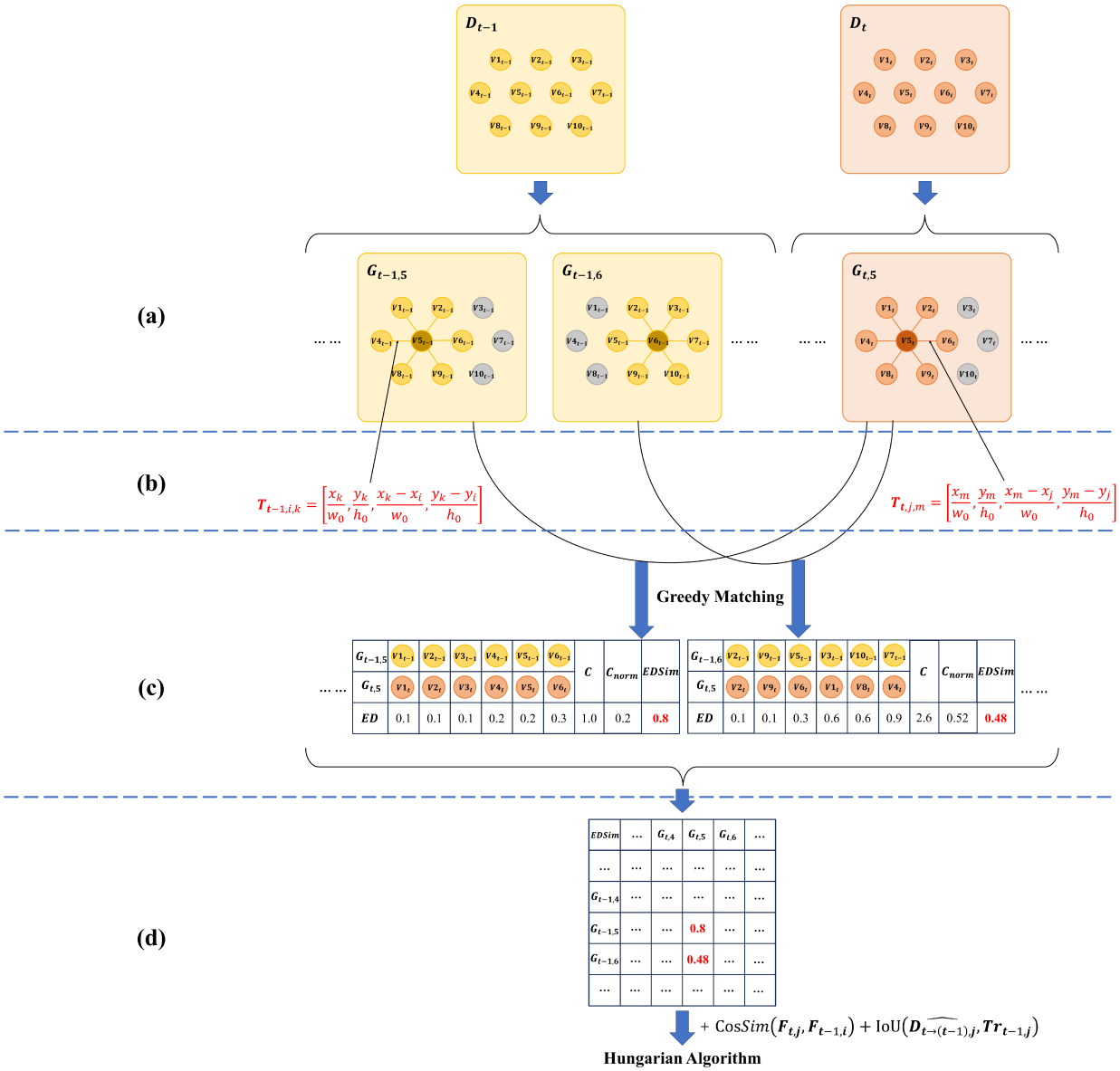


Fig. 6. Interframe target association based on JFM. (a) Subgraph construction. (b) Topological relationship description. (c) Topological relationship similarity measurement. (d) Interframe target association.

Among them, $\boldsymbol{GM}$ represents the output of greedy allocation, while $\mathrm{EDSim}(\boldsymbol{T}_{t,j}, \boldsymbol{T}_{t-1,i})$ denotes the topological similarity between the target $\boldsymbol{V}_{t,j}$ and the target $\boldsymbol{V}_{t-1,i}$.

2) Robust Interframe Association Between Targets Based on Joint Feature Similarity Measurement: After obtaining the detection results of two adjacent frames, the DLA-34 appearance features $\boldsymbol{F}_{t-1}$ and $\boldsymbol{F}_t$ and the motion offset $\boldsymbol{Disp}_{t-1}^{t}$ at the corresponding positions can be easily obtained. Therefore, the similarity of individual features between frames can be evaluated, as shown in Fig. 6(d). This evaluation includes appearance features, motion offsets, and topological structures to enhance the accuracy of interframe association. The similarity of appearance features is calculated using cosine similarity, denoted as CosSim. The similarity of motion offsets $\mathrm{IoU}(\boldsymbol{B}_{t,j}, \widehat{\boldsymbol{B}_{t,i}})$ is evaluated by the intersection over union (IoU) of the mapping value $\boldsymbol{D}_{\widehat{t\to(t-1),j}}$ of the detection box $\boldsymbol{D}_{t,j}$ of the tth frame in the $(t-1)$th frame with the tracking box $\boldsymbol{Tr}_{t-1,j}$ of the $(t-1)$th frame. The calculation process of $\boldsymbol{D}_{\widehat{t\to(t-1),j}}$ is shown in (19). The similarity of the topological structure is represented as $\mathrm{Sim}(\boldsymbol{T}_{t,j}, \boldsymbol{T}_{t-1,i})$, obtained from (18). Finally, the overall similarity of individual features $\mathrm{Sim}(\boldsymbol{V}_{t,j}, \boldsymbol{V}_{t-1,i})$ is expressed by (20)

$$\boldsymbol{D}_{\widehat{t\to(t-1),j}} = \boldsymbol{D}_{t,j} + \boldsymbol{Disp}_{t-1}^{t} \tag{19}$$

$$\begin{aligned}\mathrm{Sim}\left(\boldsymbol{V}_{t,j}, \boldsymbol{V}_{t-1,i}\right) = {} & \mathrm{CosSim}\left(\boldsymbol{F}_{t,j}, \boldsymbol{F}_{t-1,i}\right) \\ & + \mathrm{IoU}\left(\boldsymbol{D}_{\widehat{t\to(t-1),j}}, \boldsymbol{Tr}_{t-1,j}\right) \\ & + \mathrm{EDSim}\left(\boldsymbol{T}_{t,j}, \boldsymbol{T}_{t-1,i}\right).\end{aligned} \tag{20}$$

When $\mathrm{Sim}(\boldsymbol{V}_{t,j}, \boldsymbol{V}_{t-1,i})$ is larger, the individual features of $\boldsymbol{V}_{t,j}$ and $\boldsymbol{V}_{t-1,i}$ are more similar, indicating a higher probability that they represent the same target. Based on $\mathrm{Sim}(\boldsymbol{V}_{t,j}, \boldsymbol{V}_{t-1,i})$, the interframe association result of the tth frame can be obtained by the HA.

The pseudocode of MTT-STPTR is presented in Algorithm 1.

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Algorithm 1 MTT-STPTR Algorithm

Input: the current image $\boldsymbol{I_t}$, the previous image $\boldsymbol{I_{t-1}}$, the previous detection bounding boxes $\boldsymbol{D_{t-1}}$, the previous tracking bounding boxes $\boldsymbol{Tr_{t-1}}$, maximum threshold of neighbor target number Thr_{TMN}.

Output: the current detection bounding boxes $\boldsymbol{D_t}$, the current tracking bounding boxes $\boldsymbol{Tr_t}$.

repeat
  # 1. Extract the appearance features
  Extract the features $\boldsymbol{F_t}$ and $\boldsymbol{F_{t-1}}$ from the image $\boldsymbol{I_t}$ and $\boldsymbol{I_{t-1}}$ using DLA-34;
  # 2. Feature enhancement based on STRSA
  Construct the undirected bipartite graph $\widetilde{\boldsymbol{G}}^{\boldsymbol{t}}_{\boldsymbol{t-1}}$ and $\boldsymbol{V_{t-1}}$ using STSA and the cost volume $\widetilde{\boldsymbol{V}}_{\boldsymbol{t}}$ by Eq. (6)–Eq. (7);
  Obtain enhanced feature maps $\boldsymbol{F}^*_{\boldsymbol{t-1}}$ and $\boldsymbol{F}^*_{\boldsymbol{t}}$ through graph attention aggregation network by Eq. (8)–Eq. (10);
  # 3. Extract the motion clues with MGSTCL
  Calculate the embedding vectors $\boldsymbol{E_t}$ and $\boldsymbol{E_{t-1}}$ using MGSTCL and the cost volume $\boldsymbol{CV}$ by Eq. (1)–Eq. (2);
  Find the maximum response in $\boldsymbol{E_{t-1}}$ for each pixel in $\boldsymbol{E_t}$ with $\boldsymbol{CV}$ and form the offset matrix $\boldsymbol{Disp^t_{t-1}}$;
  # 4. Detect targets
  Calculate the appearance features enhanced by motor cues $\boldsymbol{F_t}'$ by Eq. (3)–Eq. (5);
  Calculate the current detection bounding boxes $\boldsymbol{D_t}$ by regression using the detection head.
  # 5. Associate the inter-frame objects based on JFM
  Calculate the topological similarity $EDSim(\boldsymbol{T_{t,j}}, \boldsymbol{T_{t-1,i}})$ of inter-frame targets by Eq. (16)–Eq. (18);
  Calculate the similarity of inter-frame targets $Sim(\boldsymbol{V_{t,j}}, \boldsymbol{V_{t-1,i}})$ based on joint features including appearance, motion and topology by Eq. (20);
  Associate inter-frame targets based on $Sim(\boldsymbol{V_{t,j}}, \boldsymbol{V_{t-1,i}})$ using the HA;
until the end of video sequence.

TABLE I
EXPERIMENT PARAMETERS

Hyper-parameters	Value
Learning Rate	1.25e-4
Epochs	50
Batch Size	4
Terminal Threshold Thr_{TMN}	30

IV. EXPERIMENT

A. Dataset and Experimental Details

Two large-scale satellite-video multi-target tracking datasets, the VISO dataset [48] and the AIR-MOT dataset [20], are used in the experiment. The videos are mainly from the Jilin-1 satellite.

1) VISO Dataset: The VISO dataset is released by National University of Defense Technology, China. The annotated moving targets only include vehicles. The dataset contains 47 videos, each with a size of 1024 × 1024 pixels, a resolution of 0.92 m, and 324–326 frames. Since each vehicle in this dataset only occupies a dozen or even a few pixels, which are extremely weak and small targets, multi-target tracking is very challenging.

2) AIR-MOT Dataset: The AIR-MOT dataset is released by Aerospace Information Research Institute of Chinese Academy of Sciences. It contains two types of annotated moving targets: aircraft and ships. The dataset contains 80 public videos, with 29 videos annotated for aircraft and 30 for ships, and the remaining videos have no annotated targets. The size of all videos is 1920 × 1080 pixels, the resolution is 0.91–1.27 m, and the number of frames is 70–326 frames. The dataset presents challenges such as complex backgrounds, motion blur, and scale variations.

After excluding videos without targets, the remaining videos were uniformly divided into subvideos with a size of 1024 × 1024 pixels, and each dataset was divided into the training dataset and the test dataset in a ratio of 8:2 for the quantitative and qualitative experiments.

The experimental environment was the Ubuntu 20.04 operating system and PyTorch. The Adam parameter optimization method was used to complete the model training on an NVIDIA GeForce RTX 3090 graphics card. Some of the important hyper-parameter settings are listed in Table I.

B. Evaluation Indices

Multiobject-tracking accuracy (MOTA) and identification $F1$-score (IDF1) are two commonly used indicators to evaluate the accuracy of MTT. MOTA comprehensively considers various error situations in the tracking process, including false positives, false negatives, and the number of mismatches, which can be expressed as

$$\text{MOTA} = 1 - \frac{(\text{FN} + \text{FP} + \text{IDSW})}{\text{GT}} \tag{21}$$

where FN indicates the number of missing targets in the video, FP is the number of false targets, and identity switching (IDSW) is the number of incorrectly associated targets.

Compared with MOTA, which depends on the detection accuracy of multi-target trackers, IDF1 focuses on the continuity of the trajectory. IDF1 is the harmonic mean of identification precision (IDP) and identification recall (IDR), which can be expressed as

$$\text{IDF1} = \frac{2 \times \text{IDP} \times \text{IDR}}{\text{IDP} + \text{IDR}}. \tag{22}$$

In addition, frames per second (FPS) is used to measure the real-time processing capability of the multi-target tracking method.

V. RESULTS AND DISCUSSION

A. Ablation Experiment

In order to verify the effectiveness of the proposed STRSA, MGSTCL, and JFM in MTT-STPTR, ablation experiments

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TABLE II
PERFORMANCE OF MODELS IN ABLATION EXPERIMENTS ON THE VISO DATASET

ID	Model	$MOTA(\%)\uparrow$	$IDF1(\%)\uparrow$	GT	$FP\downarrow$	$FN\downarrow$	$IDSW\downarrow$	$IDP(\%)\uparrow$	$IDR(\%)\uparrow$	$FPS\uparrow$
1	Baseline	62.50	73.50	140122	771	50862	896	93.90	60.40	6.52
2	Baseline-STRSA	65.40	79.70	140122	27564	19013	1916	77.40	82.10	**6.94**
3	Baseline-STRSA-MGSTCL	65.80	78.40	140122	21595	24434	1926	79.20	77.60	3.52
4	Baseline-STRSA-MGSTCL-JFM(Ours)	**66.80**	**81.20**	140122	21595	24434	445	82.00	80.40	2.43

TABLE III
PERFORMANCE OF MODELS IN ABLATION EXPERIMENTS ON THE AIR-MOT DATASET

ID	Model	$MOTA(\%)\uparrow$	$IDF1(\%)\uparrow$	GT	$FP\downarrow$	$FN\downarrow$	$IDSW\downarrow$	$IDP(\%)\uparrow$	$IDR(\%)\uparrow$	$FPS\uparrow$
1	Baseline	35.20	49.90	184080	6753	111939	564	83.10	35.60	**7.34**
2	Baseline-STRSA	42.60	59.90	184080	3162	102059	369	94.70	43.80	6.42
3	Baseline-STRSA-MGSTCL	51.90	68.10	184080	6026	81423	1010	91.80	54.20	6.57
4	Baseline-STRSA-MGSTCL-JFM(Ours)	**52.40**	**69.50**	184080	6026	81418	98	93.60	55.30	5.68

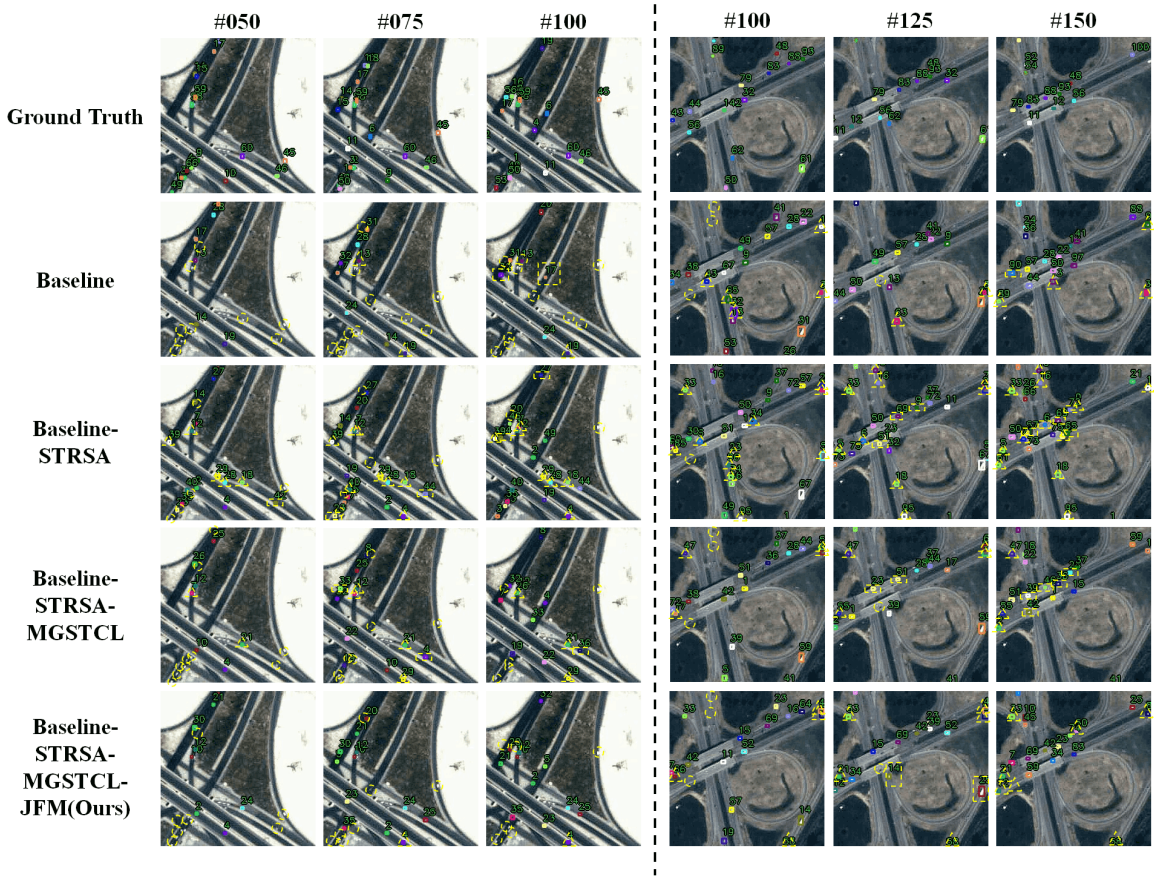


Fig. 7. Visualization of the results of each experiment in the ablation experiment on the VISO dataset.

are performed in this section. Tables II and III show the contribution of each module in this method. The tracking results of the frame sequences are visualized in Fig. 7, where different individuals are represented by solid boxes of different colors and the numbers above them represent the individual numbers. Yellow dotted circles represent FNs, yellow dotted triangles represent FPs, and yellow dotted boxes represent IDSWs. In addition, Fig. 8 shows the target trajectories of each experiment in the ablation experiment, which contains three coordinate axes: X coordinate, Y coordinate, and frame number of frame sequences.

From Tables II and III, with the gradual accumulation of the three improved modules, the accuracy has been significantly improved although the efficiency has been decreased. Specifically, compared with the baseline method, MTT-STPTR improved MOTA and IDF1 by 4.30% and 7.70%, respectively, on the VISO dataset and improved by 17.20% and 19.60%, respectively, on the AIR-MOT dataset. It is worth noting that MTT-STPTR achieves the lowest IDSW in both datasets, which can greatly improve the continuity of target trajectories.

1) Effectiveness of STRSA: From Tables II and III, STRSA improves MOTA and IDF1 by 2.90% and 6.20% on the VISO

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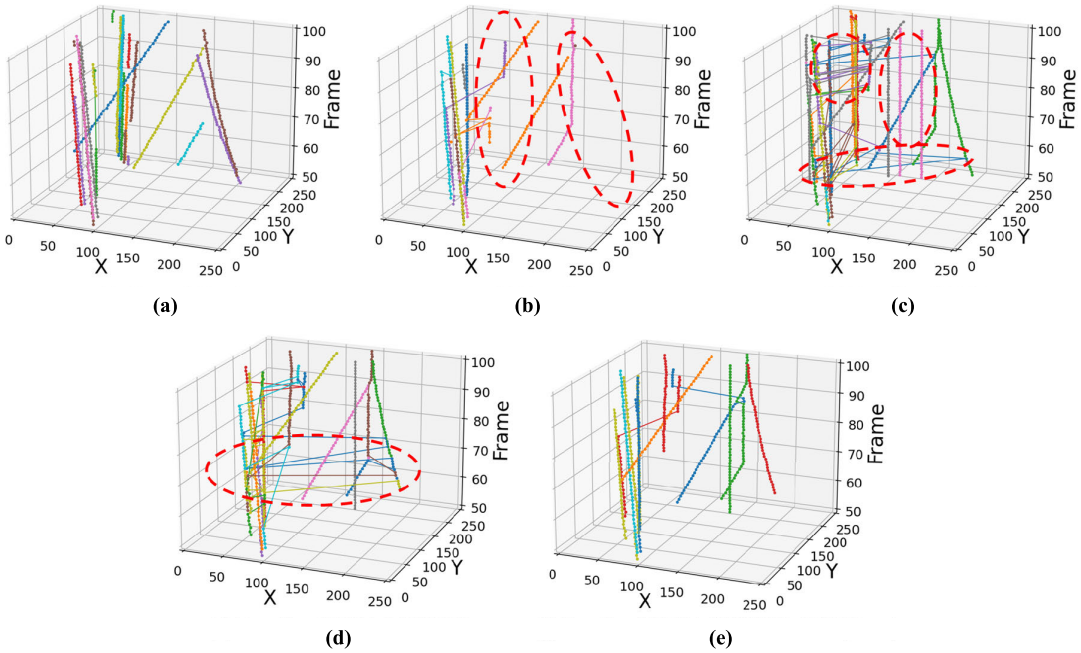


Fig. 8. Trajectories of each experiment in the ablation experiment. (a) Ground truth. (b) Baseline. (c) Baseline-STRSA. (d) Baseline-STRSA-MGSTCL. (e) Baseline-STRSA-MGSTCL-JFM (Ours).

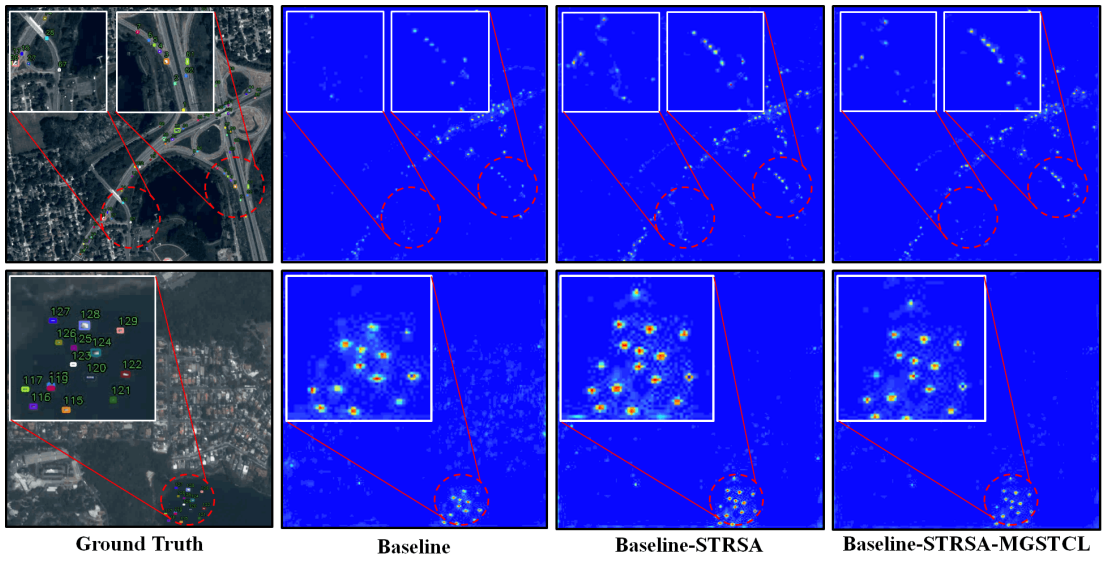


Fig. 9. Visualization of the feature maps of each experiment in the ablation experiment.

dataset and by 7.40% and 10.00% on the AIR-MOT dataset. The improvement is attributed to STRSA's ability to accurately compensate for historical information through a local search strategy, which effectively enhances feature representation and detection performance for small targets (see Fig. 9). In addition, FN is reduced by 31 849 and 9880 on VISO and AIR-MOT datasets, respectively. From the sequence images of Figs. 7 and 8(a)–(c), it is evident that STRSA can recall more potential small targets compared to the baseline method. Although STRSA introduces some false alarms and false associations, the cost is worthy considering the improvement in comprehensive accuracy. FP and IDSW can be reduced through subsequent improvement steps. In addition, we also analyze the performance of STRSA in the scenario where the target is occluded. As shown in Fig. 10, from the 184th frame to the 201st frame, the 25th vehicle target passes under the overpass, and the 37th vehicle target on the overpass appears in the region of interest of the 25th vehicle target. From the third row in Fig. 10, we can analyze that when the 25th vehicle target passes under the overpass in the 195th frame, STRSA does not generate a significant local area with a high response value like the 184th and 190th frames. Therefore, no new trajectory is generated for the 25th vehicle target, and it is not confused with the 37th vehicle target.

2) Effectiveness of MGSTCL: From Tables II and III, MGSTCL improves MOTA and IDF1 by 9.30% and 8.20%

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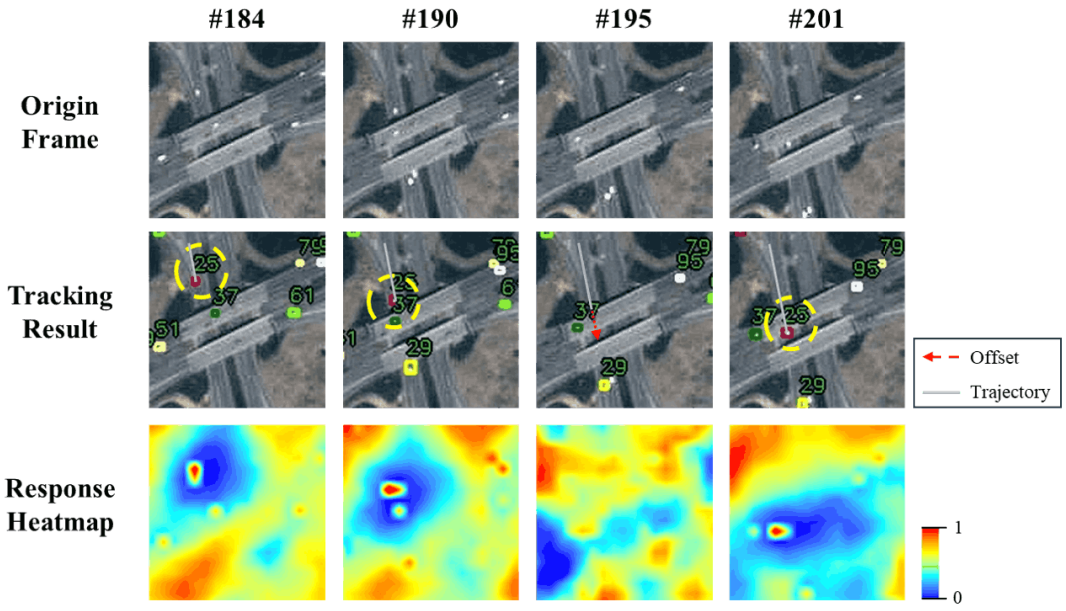


Fig. 10. Performance of MTT-STPTR in scenes where the target reappears after being occluded.

on the AIR-MOT dataset, respectively. The improvement is attributed to MGSTCL's introduction of a spatial-temporal comparative learning method at the individual-category level, which effectively balances performance between detection and association, thereby enhancing overall accuracy. On the VISO dataset, MGSTCL shows more modest results, with MOTA improved by only 0.40%, while IDF1 decreases by 1.30%. The limitation arises because the VISO dataset only contains one type of target and vehicle, and MGSTCL does not show its advantages. However, from Fig. 9, whether on the VISO dataset or the AIR-MOT dataset, Baseline-STRSA-MGSTCL can enhance the feature representation of small targets while reducing the interference of surrounding noise features, making the boundaries between targets clearer and facilitating the extraction of cluster targets.

3) Effectiveness of JFM: From Tables II and III, JFM improves MOTA and IDF1 by 1.00% and 2.80% on the VISO dataset and by 0.50% and 1.40% on the AIR-MOT dataset. The improvement is due to JFM's construction of a joint feature similarity measurement method that incorporates appearance, motion, and topology. The approach enhances the identifiability of target individuals in dense scenes, thereby improving the accuracy of interframe associations. Therefore, the number of IDSW is reduced by 1487 and 836 on VISO and AIR-MOT datasets, respectively. From Fig. 8, JFM results in fewer false associations and enhances the integrity and continuity of the trajectories compared to the baseline method. In addition, JFM can also be applied in reidentification scenarios. As shown in Fig. 10, the interrupted trajectory of the 25th vehicle target is continuously extrapolated to obtain the position of the current frame according to the previous offset, until the interruption time exceeds 30 frames. When the 25th vehicle target drives out from under the overpass, a multidimensional feature similarity measurement is conducted with the extrapolated virtual target using the JFM module in the 201st frame to achieve reidentification of the occluded target.

In summary, through the comprehensive utilization of the three modules of STRSA, MGSTCL, and JFM, MTT-STPTR can utilize the spatial-temporal and topological relationships between targets for feature propagation and enhancement, as well as individual feature modeling, and improve the tracking accuracy of small and dense targets in satellite videos.

B. Comparative Experiments

To verify the advantages of MTT-STPTR in satellite videos, nine representative general multi-target tracking methods mentioned in Section I, namely, CenterTrack [18], FairMOT [16], PermaTrack [19], TraDeS [7], ByteTrack [31], QDTrack [49], MOTRv2 [50], Path-Consistency [51], and TLTD-MOT [52], and three satellite-video-specific multi-target tracking methods, namely, CKDNet + SMTNet [2], TGraM [20], and CFTracker [24], were selected for comparison. Tables IV and V present the accuracy of different methods.

As shown in Tables IV and V, the MOTA and IDF1 of MTT-STPTR on the VISO dataset are 66.80% and 81.20%, and the MOTA and IDF1 on the AIR-MOT dataset are 52.40% and 69.50%, respectively. Compared with other methods, MTT-STPTR has achieved SOTA, with optimal tracking accuracy and stability for multi-targets in satellite videos.

1) Performance on the VISO Dataset: As shown in Table IV, MOTRv2, FairMOT, and CenterTrack exhibit poor performance, while TraDeS, CFTracker, and MTT-STPTR demonstrate significantly better results.

MOTRv2 follows the TBD design principle, generating learnable and propagable feature vectors based on detection results from YOLOX, and performs interframe target matching

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TABLE IV
PERFORMANCE OF MODELS IN COMPARATIVE EXPERIMENT ON THE VISO DATASET

ID	Model	$MOTA$(%)↑	$IDF1$(%)↑	GT	$FP\downarrow$	$FN\downarrow$	$IDs\downarrow$	IDP(%)↑	IDR(%)↑	$FPS\uparrow$
1	(ECCV20)CenterTrack	2.30	4.70	140122	0	136557	287	95.10	2.40	2.07
2	(IJCV21)FairMOT	1.10	2.20	140122	0	138545	14	100.00	1.10	**19.71**
3	(ICCV21)PermaTrack	26.90	41.90	140122	0	101590	885	97.00	26.70	8.28
4	(CVPR21)TraDeS	62.50	73.50	140122	771	50862	896	93.90	60.40	6.52
5	(ISPRS P&RS21)CKDNet+SMTNet	17.60	43.70	140122	23214	92099	172	64.80	32.90	13.91
6	(IEEE TGRS22)TGraM	1.10	2.30	140122	0	138516	41	100.00	1.10	6.55
7	(ECCV22)ByteTrack	16.90	46.00	140122	30093	86226	174	61.40	36.80	11.75
8	(TPAMI23)QDTrack	1.10	2.20	140122	0	138583	22	100.00	1.10	15.64
9	(CVPR23)MOTRv2	0.10	0.30	140122	0	139945	5	100.00	0.10	12.19
10	(IEEE TGRS23)CFTracker	66.40	61.90	140122	20838	19366	6833	61.50	62.20	1.98
11	(CVPR24)Path-Consistency	53.80	68.60	140122	1431	36984	384	93.80	54.10	10.13
12	(CVPR24)TLTD-MOT	0.80	1.80	140122	0	138851	143	100.00	0.90	7.36
13	(Ours)MTT-STPTR	**66.80**	**81.20**	140122	21595	24434	445	82.00	80.40	2.43

TABLE V
PERFORMANCE OF MODELS IN COMPARATIVE EXPERIMENT ON THE AIR-MOT DATASET

ID	Model	$MOTA$(%)↑	$IDF1$(%)↑	GT	$FP\downarrow$	$FN\downarrow$	$IDSW\downarrow$	IDP(%)↑	IDR(%)↑	$FPS\uparrow$
1	(ECCV20)CenterTrack	38.70	55.70	184080	7259	104630	1011	86.90	40.90	2.59
2	(IJCV21)FairMOT	9.20	16.90	184080	37	166909	164	98.80	9.20	23.49
3	(ICCV21)PermaTrack	10.20	18.40	184080	37	165131	52	98.20	10.10	7.14
4	(CVPR21)TraDeS	35.20	49.90	184080	6753	111939	564	83.10	35.60	7.34
5	(ISPRS P&RS21)CKDNet+SMTNet	21.00	39.10	184080	9443	134348	63	81.80	25.70	16.54
6	(IEEE TGRS22)TGraM	7.80	14.80	184080	232	169270	143	98.00	8.00	**43.42**
7	(ECCV22)ByteTrack	21.30	40.60	184080	11545	132951	69	80.10	27.20	15.14
8	(TPAMI23)QDTrack	6.00	11.50	184080	232	172324	413	93.80	6.10	53.48
9	(CVPR23)MOTRv2	46.00	68.30	184080	30738	68370	337	77.00	61.30	11.96
10	(IEEE TGRS23)CFTracker	49.70	65.40	184080	2989	88317	1320	93.60	50.20	2.59
11	(CVPR24)Path-Consistency	33.40	50.00	184080	510	120625	46	98.50	33.50	13.78
12	(CVPR24)TLTD-MOT	17.80	29.60	184080	303	150599	476	95.40	17.50	24.62
13	(Ours)MTT-STPTR	**52.40**	**69.50**	184080	6026	81418	98	93.60	55.30	5.68

through Transformer. However, the Transformer suffers from high computational complexity and slow convergence and performs poorly on small-sized targets with scarce texture features, resulting in the highest FN and lowest MOTA and IDF1, which are 139 945, 0.10%, and 0.30%, respectively. FairMOT can simultaneously output the detection bounding boxes and the appearance feature vectors for interframe target association, thereby improving processing efficiency. However, it relies solely on single-frame images for feature extraction, which hinders its ability to distinguish between individuals with scarce texture features in satellite videos, resulting in higher FN and lower MOTA and IDF1, which are 138 545, 1.10%, and 2.20%, respectively. In contrast, CenterTrack and Path-Consistency utilize multiframe information. CenterTrack belongs to the JDT paradigm and introduces multiframe information in feature extraction to enhance feature expression and generate offsets of targets. However, it mainly adopts the method of splicing and fusing multiframe feature maps, which lacks fine pixel-by-pixel feature alignment. Therefore, CenterTrack suffers from more missed detections during small-size target extraction, resulting in higher FN and lower IDR, which are 136 557 and 2.40%, respectively. Based on the detection results of excellent detector YOLOX [53] in ByteTrack, Path-Consistency improves multi-target tracking continuity through self-supervised cross-frame consistency learning of trajectory datasets, with significant improvements in MOTA and IDF1, 36.90% and 22.60%, respectively, which is highly competitive in terms of both accuracy and efficiency.

In order to further achieve effective compensation of historical information and enhance target features, TraDeS and CFTracker were proposed. TraDeS aggregates the historical frame motion information related to the target using offsets and achieves target tracking while strengthening the detection results, thereby improving the efficiency of feature utilization. TraDeS achieves high MOTA and IDF1, which are 62.50% and 73.50%, respectively. Building on CenterTrack, CFTracker enhances detection results of the current frame through offset and cross-attention mechanisms, thereby further improving the accuracy of multi-target tracking. Its MOTA and IDF1 are as high as 66.40% and 61.90%, respectively. However, they mainly use the similarity of motion features as clues for interframe association, which can lead to failures in local scenes with densely distributed vehicles, resulting in suboptimal tracking performance. The highest IDSWs of TraDeS and CFTracker are 896 and 6833, respectively. In contrast, IDSWs of MTT-STPTR are reduced by 451 and 6388, and IDF1 is improved by 7.70% and 19.30%. Compared to the FPS of MTT-STPTR, which is 2.43, TraDeS achieves a higher FPS of 6.52. The difference can be attributed to TraDeS having more FNs in its target detection results than MTT-STPTR. Consequently, the number of targets that need to be associated between frames is lower, which reduces the processing time required for each frame. In contrast, MTT-STPTR incorporates topological relationships through the JFM module to enrich the feature descriptions of individual targets and improve the distinguishability of features among different individuals. This

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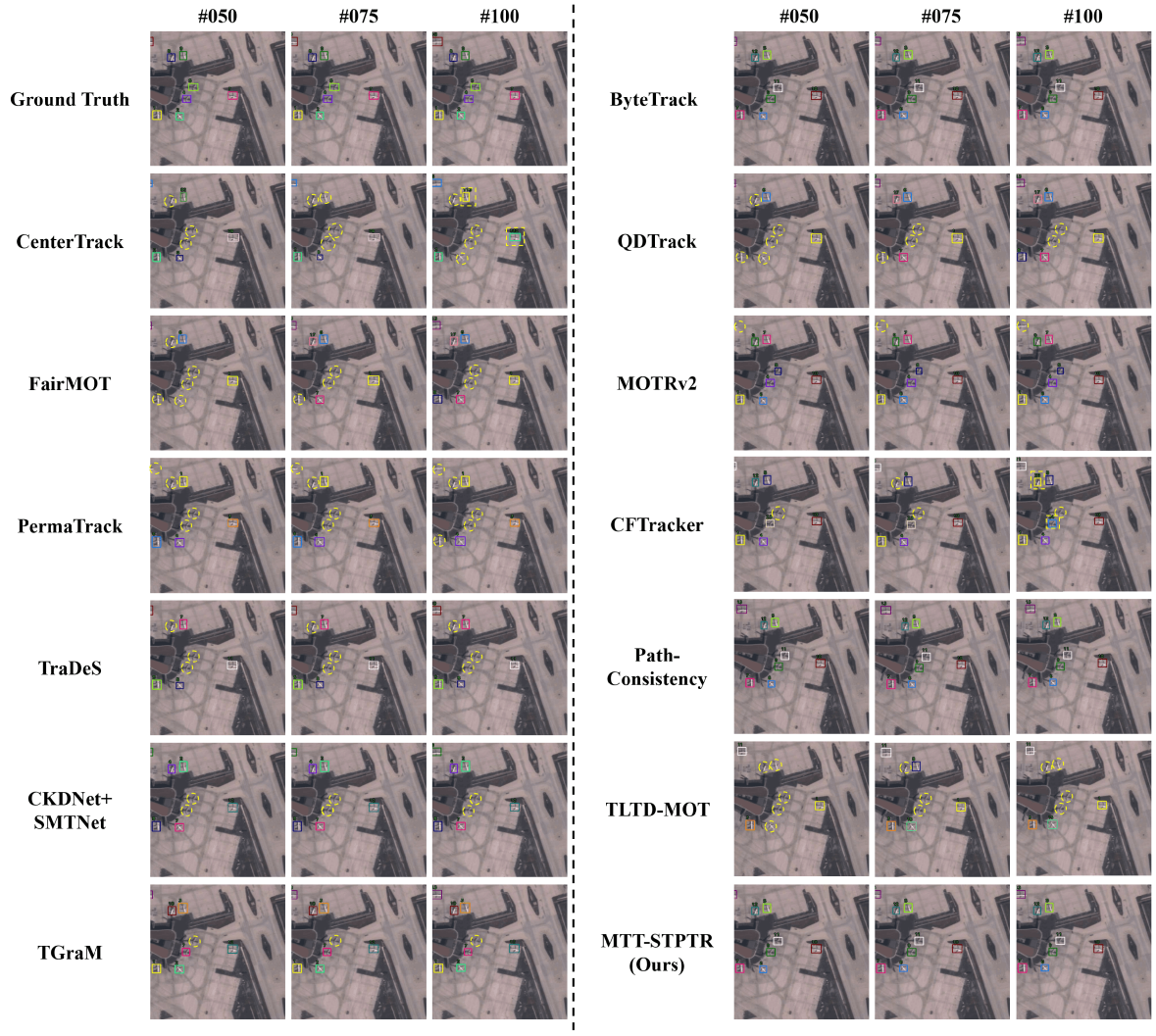


Fig. 11. Visualization of the results of each experiment in the comparative experiment on the AIR-MOT dataset (plane).

approach enables comprehensive extraction and long-term tracking of dense and small vehicles in satellite videos.

2) Performance on the AIR-MOT Dataset: As shown in Table V, QDTrack, PermaTrack, and ByteTrack exhibit poor performance, whereas MOTRv2, CFTracker, and MTT-STPTR demonstrate significantly better results.

QDTrack introduces contrastive learning into the multi-target tracking task, which is used to enhance the appearance features of individuals, thereby improving the accuracy of interframe association. However, targets observed from a long-distance bird's-eye view in satellite videos often lack texture and perspective features, making it challenging to optimize this contrastive learning approach, resulting in higher FN and lower IDR, which are 172 324 and 6.10%, respectively. From the sequence images of Figs. 11 and 12, QDTrack has many missed targets. In addition, Fig. 13(i) also has many interrupted tracks, indicating that QDTrack is inefficient when relying solely on appearance features. Based on CenterTrack, PermaTrack introduces gated recurrent units to selectively memorize or forget the feature information from historical frames, thereby generating more accurate detection results and offsets. However, as shown in Table V, PermaTrack's performance fell short of expectations, with low MOTA and IDF1 of 10.20% and 18.40%, respectively, and high FN of 165 131. Many lost tracks can be observed in Fig. 13(d). This may be due to PermaTrack's use of all feature information including background and target, which is susceptible to interference from the complex backgrounds present in satellite videos during training. ByteTrack is based on the design principle of TBD and incorporates the low-confidence detection box recall strategy in interframe association, improving trajectory continuity. Therefore, it has a higher IDF1 and lower IDSW and FN, which are 40.60%, 69, and 132 951, respectively. However, its performance heavily relies on the high-precision single-frame detector YOLOX, requiring multiple adjustments to the confidence threshold, resulting in low accuracy, and its MOTA is only 21.30%. In addition, based on FairMOT, TLTD-MOT alleviates the impact of long-tail distribution of trajectories on multi-target tracking performance by grouping trajectory datasets for training. Its MOTA and IDF1 are improved by 8.60% and 12.70%, respectively. However, they

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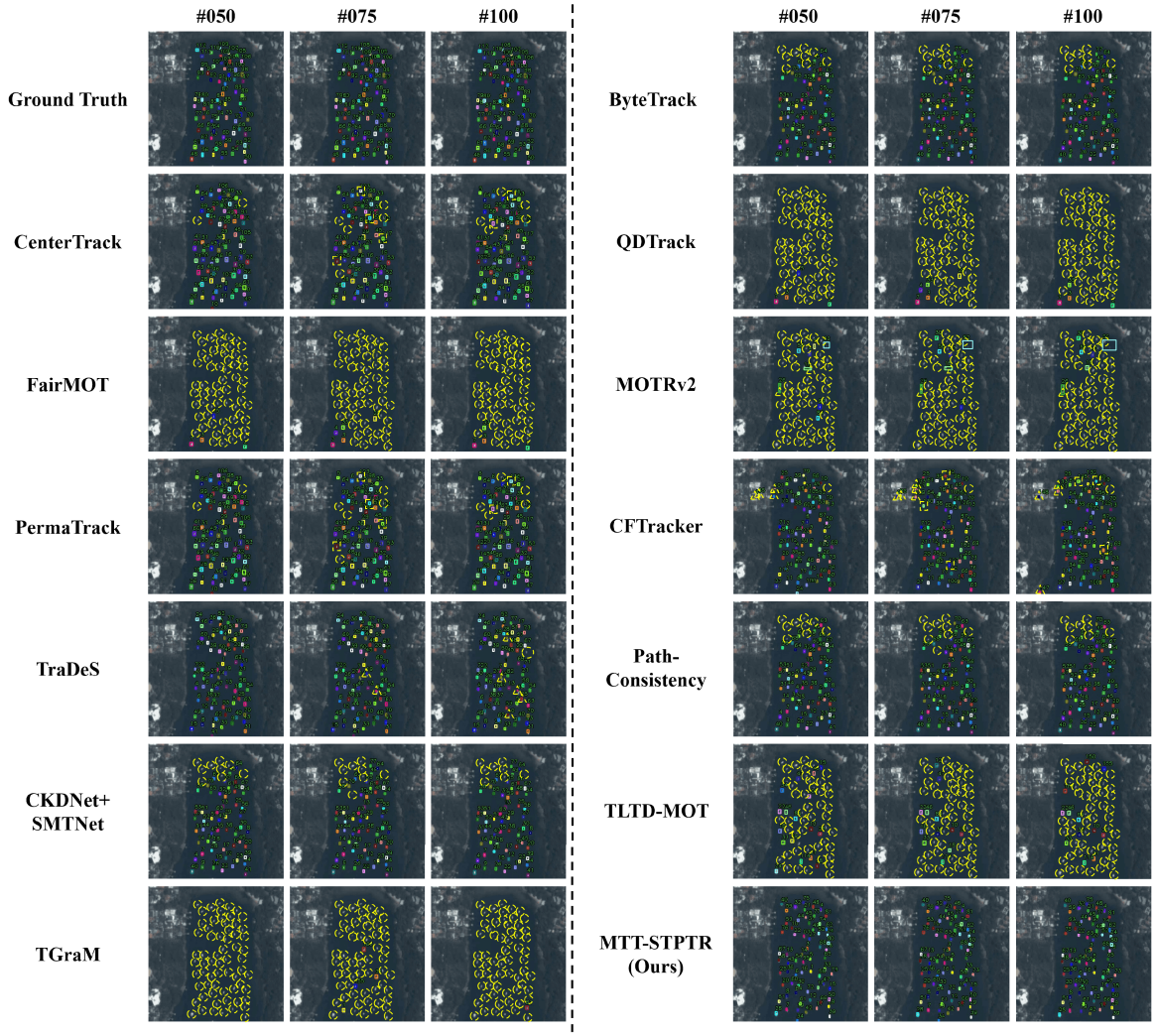


Fig. 12. Visualization of the results of each experiment in the comparative experiment on the AIR-MOT dataset (ship).

rely on features from single-frame images, making it difficult to distinguish between individuals and, thereby, limiting further improvements in accuracy.

Compared with the VISO dataset, the AIR-MOT dataset mainly includes larger targets such as airplanes and ships, so MOTRv2 shows higher MOTA and IDF1, which are 46.00% and 68.30%, respectively. The detection of large-sized ship targets in Fig. 12 shows that MOTRv2 has a strong ability to model long-distance spatial features. There are fewer interrupted trajectories in Fig. 13(j), which also highlights the advantages of MOTRv2 in long-term tracking tasks. In addition, MOTRv2 also has a higher processing efficiency, with an FPS of 11.96. CFTracker still maintains stable and excellent performance in the AIR-MOT dataset, with a high MOTA and IDF1 of 49.70% and 65.40%, respectively. The success is attributed to its time-series feature enhancement and cross-frame consistency training methods, showing lower FP and FN of 2989 and 88 317, respectively. However, the interframe association method based on offset matching leads to a higher IDSW of 1320, with frequent trajectory interruptions observed in Fig. 13(k). Compared with previous methods, MTT-STPTR utilizes the implicit target relationship priors in satellite videos. By utilizing spatial-temporal relationship priors between targets, it incorporates an STRSA mechanism and multigranularity contrast learning to enhance target features. The topological relationships between targets further improve the distinguishability of individual features. The design makes MTT-STPTR particularly suitable for multi-target tracking tasks in satellite videos, achieving optimal speed and efficiency in both dense and sparse target scenarios.

C. Parameter Sensitivity Analysis

The proposed JFM introduces the upper limit number of neighbor targets $\mathrm{Thr}_{\mathrm{NTN}}$ and multiple association cues, whose settings have an important impact on the performance of multi-target tracking. Therefore, this section mainly conducts sensitivity analysis on the hyperparameter $\mathrm{Thr}_{\mathrm{NTN}}$ and different association cue combinations. In addition, this section will also demonstrate the processing efficiency of our proposed method MTT-STPTR under different image resolutions and target numbers.

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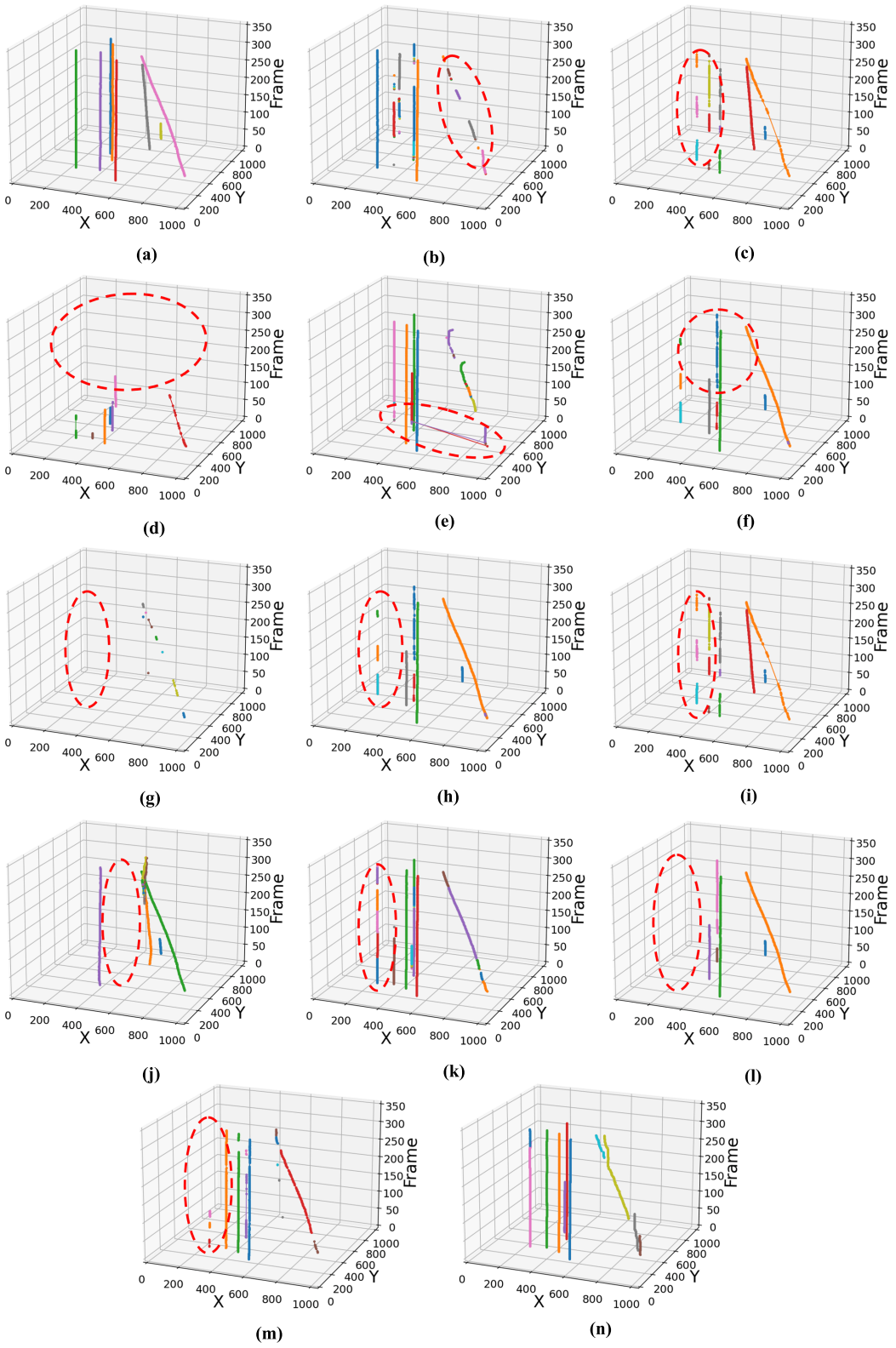


Fig. 13. Trajectories of each experiment in the comparative experiment. (a) Ground truth. (b) CenterTrack. (c) FairMOT. (d) PermaTrack. (e) TraDeS. (f) CKDNet + SMTNet. (g) TGraM. (h) ByteTrack. (i) QDTrack. (j) MOTRv2. (k) CFTracker. (l) Path-Consistency. (m) TLTD-MOT. (n) MTT-STPTR.

1) Upper Limit Number of Neighbor Targets Thr_{NTN}: When Thr_{NTN} is larger, the number of neighbor targets participating in describing the topological relationship of each target is greater, and the individual features are more distinguishable, but the efficiency of processing will decrease. As shown in Table VI, for the VISO dataset, when $\text{Thr}_{\text{NTN}} = 5, 10, 15,$

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TABLE VI

QUANTITATIVE RESULTS WITH DIFFERENT MAXIMUM THRESHOLDS OF NEIGHBOR TARGET NUMBER THR_{NTN} ON THE VISO DATASET

ID	Thr_{NTN}	$MOTA(\%)\uparrow$	$IDF1(\%)\uparrow$	GT	$FP\downarrow$	$FN\downarrow$	$IDSW\downarrow$	$IDP(\%)\uparrow$	$IDR(\%)\uparrow$	$FPS\uparrow$
1	5	**66.80**	**81.20**	140122	21595	24434	445	82.00	80.40	**2.43**
2	10	**66.80**	**81.20**	140122	21595	24434	424	82.10	80.40	2.01
3	15	**66.80**	**81.20**	140122	21595	24435	425	82.10	80.40	1.63
4	20	**66.80**	81.10	140122	21595	24434	439	82.00	80.30	1.39
5	25	**66.80**	81.00	140122	21595	24434	432	81.90	80.20	1.15

TABLE VII

QUANTITATIVE RESULTS WITH DIFFERENT MAXIMUM THRESHOLDS OF NEIGHBOR TARGET NUMBER THR_{NTN} ON THE AIR-MOT DATASET

ID	Thr_{NTN}	$MOTA(\%)\uparrow$	$IDF1(\%)\uparrow$	GT	$FP\downarrow$	$FN\downarrow$	$IDSW\downarrow$	$IDP(\%)\uparrow$	$IDR(\%)\uparrow$	$FPS\uparrow$
1	5	**52.40**	**69.50**	184080	6026	81418	98	93.60	55.30	**5.68**
2	10	**52.40**	**69.50**	184080	6026	81423	104	93.70	55.30	5.25
3	15	**52.40**	**69.50**	184080	6026	81423	102	93.70	55.30	4.85
4	20	**52.40**	**69.50**	184080	6026	81423	102	93.70	55.30	4.46
5	25	**52.40**	**69.50**	184080	6026	81418	102	93.70	55.30	4.06

TABLE VIII

QUANTITATIVE RESULTS WITH DIFFERENT WAYS OF INTERFRAME TARGET ASSOCIATION ON THE VISO DATASET

ID	Ways	$MOTA(\%)\uparrow$	$IDF1(\%)\uparrow$	GT	$FP\downarrow$	$FN\downarrow$	$IDSW\downarrow$	$IDP(\%)\uparrow$	$IDR(\%)\uparrow$	$FPS\uparrow$
1	Motion	65.80	78.40	140122	21595	24434	1926	79.20	77.60	**4.08**
2	Appearance	65.70	77.80	140122	21595	24434	2038	78.60	77.00	3.59
3	Topology	**66.80**	80.00	140122	21595	24433	431	80.90	79.20	2.67
4	Motion+Appearance	65.70	77.60	140122	21595	24433	2038	78.40	76.80	3.57
5	Appearance+Topology	**66.80**	80.60	140122	21671	24433	438	81.40	79.80	2.45
6	Motion+Topology	**66.80**	80.20	140122	21595	24434	436	81.00	79.30	2.71
7	Motion+Appearance+Topology	**66.80**	**81.20**	140122	21595	24434	445	82.00	80.40	2.43

TABLE IX

QUANTITATIVE RESULTS WITH DIFFERENT WAYS OF INTERFRAME TARGET ASSOCIATION ON THE AIR-MOT DATASET

ID	Ways	$MOTA(\%)\uparrow$	$IDF1(\%)\uparrow$	GT	$FP\downarrow$	$FN\downarrow$	$IDSW\downarrow$	$IDP(\%)\uparrow$	$IDR(\%)\uparrow$	$FPS\uparrow$
1	Motion	51.90	68.10	184080	6026	81423	1010	91.80	54.20	**6.80**
2	Appearance	51.90	67.80	184080	6026	81423	1037	91.30	53.90	6.50
3	Topology	52.40	68.90	184080	6026	81423	103	92.80	54.80	5.89
4	Motion+Appearance	51.90	67.70	184080	6026	81423	1141	91.20	53.90	6.53
5	Appearance+Topology	52.40	69.00	184080	6026	81423	106	92.90	54.90	5.70
6	Motion+Topology	52.40	69.50	184080	6026	81423	116	93.60	55.30	5.88
7	Motion+Appearance+Topology	**52.40**	**69.50**	184080	6026	81418	98	93.60	55.30	5.68

TABLE X

PERFORMANCE OF MTT-STPTR AT DIFFERENT RESOLUTIONS AND NUMBERS OF TARGETS

Video ID	Resolution(m)	Number of Targets	$MOTA(\%)\uparrow$	$IDF1(\%)\uparrow$	$FP\downarrow$	$FN\downarrow$	$IDSW\downarrow$	$IDP(\%)\uparrow$	$IDR(\%)\uparrow$	$FPS\uparrow$
1	1.3	1691	83.50	91.00	0	279	0	100.00	83.50	**11.00**
2	1.3	21458	88.00	86.90	333	1969	278	90.40	83.50	2.81
3	1.3	34009	75.00	85.70	0	8507	3	100.00	75.00	2.06
1	1	1691	96.90	98.40	0	52	0	100.00	96.90	**7.15**
2	1	21458	88.60	89.20	482	1645	326	91.70	86.70	2.41
3	1	34009	78.60	88.10	0	7265	2	100.00	78.70	1.68
1	0.8	1691	99.90	100.00	1	0	0	99.90	100.00	**4.33**
2	0.8	21458	87.90	89.30	336	1950	305	92.90	85.90	1.72
3	0.8	34009	77.40	87.30	0	7677	4	100.00	77.40	1.23

MOTA and IDF1 reach a maximum of 66.80% and 81.20%. It indicates that JFM requires only a small number of neighbor targets to effectively model topological features, achieving good performance in densely similar multi-target tracking scenarios. When $\text{Thr}_{\text{NTN}} = 25$, IDF1 slightly decreases to 81.00%. This reduction occurs because distant targets, which

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may be located at the edges of the image, can introduce noise into the description of the target's topological relationship, leading to association errors and losses. In addition, as shown in Table VII, in the AIR-MOT dataset, when $\mathrm{Thr}_{\mathrm{NTN}}$ increases from 5 to 25, MOTA and IDF1 remain unchanged. This is because compared to the VISO dataset, the number of targets in a single video in the AIR-MOT dataset is relatively small and is less sensitive to adjustments to the upper limit of the number of neighbor targets. Therefore, we set $\mathrm{Thr}_{\mathrm{NTN}} = 5$ in the experiment to achieve optimal accuracy and efficiency simultaneously.

2) Different Combinations of Association Cues: JFM includes three similarity metrics with the same weights: motion, appearance, and topology. They can be combined in various ways to form different association cues. As shown in Tables VIII and IX, when the association cues use motion similarity or appearance similarity or motion-appearance similarity, both MOTA and IDF1 both achieve lower values on the VISO dataset and the AIR-MOT dataset, and IDSWs are higher values. It indicates that motion and appearance similarities alone are insufficient to effectively distinguish individual differences in satellite video multi-target tracking, resulting in increased association errors and losses. In contrast, when the association cues include topological similarity (Experiments 3, 5, 6, and 7), MOTA and IDF1 are improved, and IDSWs are greatly reduced. For example, on the VISO dataset, MOTA and IDF1 of appearance-topology similarity (Experiment 5) increased by 1.1% and 2.8%, respectively, compared with appearance similarity (Experiment 2); MOTA and IDF1 of motion-topology similarity (Experiment 6) increased by 1% and 1.8%, respectively, compared with motion similarity (Experiment 1); and IDSW decreased from the highest value of 2038 to the lowest value of 436. In addition, when the association cue uses motion-appearance-topology similarity, the trajectory continuity of the target performs the best. For example, on the AIR-MOT dataset, IDF1 of motion-appearance-topology similarity (Experiment 7) achieved the highest value of 69.60%, which was 0.1% and 0.5% higher than motion-topology similarity (Experiment 6) and appearance-topology similarity (Experiment 5), respectively. This demonstrates that the introduction of topological relationship description enhances the expression of individual features, and measuring the similarity of target individuals from multidimensional features helps improve the accuracy of interframe association, which reflects the significant advantages of MTT-STPTR. Furthermore, when incorporating topological similarity, FPS is slightly reduced by 0.9–1.1, indicating that the original computational efficiency advantage is largely maintained.

3) Under Different Resolutions and Numbers of Targets: Three videos with different spatial resolutions and different numbers of targets were selected for the experiment. As shown in Table X, with higher spatial resolution and more targets, FPS decreases significantly. Therefore, in practical applications, the real-time processing capability of MTT-STPTR can be improved by parallel processing of video blocks.

VI. Conclusion

This article proposes a novel MTT-STPTR, which is used for stable tracking of small and dense multi-targets in satellite videos. The rich spatial-temporal proximity and topological relationships between targets in satellite videos facilitate small target detection and accurate interframe associations. Inspired by these insights, MTT-STPTR follows the JDT paradigm and incorporates three innovative key modules: STRSA, MGSTCL, and JFM. In STRSA, a sparse spatial-temporal relationship graph is constructed based on the target regions of adjacent frames, allowing for the accurate capture of cross-frame semantics related to the targets, thereby enhancing the feature representation of small targets effectively. In MGSTCL, a new contrast loss function is designed to improve the distinguishability of target features at the individual-category granularity, thereby promoting the balance between detection and association multitask optimization, improving the accuracy of multi-target tracking. In JFM, a novel topological relationship description method is developed to construct an appearance-motion-topology joint feature representation for individuals, achieving robust tracking of clustered targets. Experiments on two public satellite-video multi-target tracking datasets, VISO and AIR-MOT, demonstrate that MTT-STPTR outperforms existing representative multi-target tracking methods in terms of MOTA and IDF1, indicating its effectiveness.

Existing deep-learning-based multi-target tracking methods for satellite videos rely heavily on large-scale annotated datasets and are typically limited to processing a small number of specific target types. For satellite videos with complex backgrounds and a rich variety of targets, these existing methods often fall short of processing requirements. Therefore, future research will focus more on multi-target tracking for satellite videos under small sample conditions and in open-world scenarios, further reducing the cost of manual annotation and enhancing the generalization capabilities of the method.

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Jianzhi Hong received the B.S. degree in geographic information science from South China Normal University, Guangzhou, China, in 2020, and the M.E. degree in resources and environment from Wuhan University, Wuhan, China, in 2022.

He is currently working toward the Ph.D. degree in photogrammetry and remote sensing with Wuhan University. His main research interests involve intelligent interpretation based on spaceborne or airborne videos, including detection, tracking and segmentation.



Taoyang Wang received the B.E. and Ph.D. degrees in photogrammetry and remote sensing from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2007 and 2012, respectively.

His doctoral dissertation was on the block adjustment of high-resolution satellite remote sensing imagery. Since 2014, he has been working with the School of Remote Sensing and Information Engineering, Wuhan University, where he became a Professor in 2023. His research interests include space photogrammetry, geometry processing of spaceborne optical/SAR/InSAR imagery, target detection, and recognition based on satellite video.



Yuqi Han received the B.Eng. degree in the field of information engineering from Beijing Institute of Technology, Beijing, China, in 2015, and the B.Sc. degree in the field of economy from the National School of Development, Peking University, Beijing, in 2015, and the Ph.D. degree from the School of Information and Electronics, Beijing Institute of Technology, in 2021.

He was a Research Fellow with the Department of Computer Science and Technology, Tsinghua University, in 2021. He is currently an Assistant Professor with the School of Information and Electronics, Beijing Institute of Technology. His research interests include computer vision, remote sensing, and unmanned aerial vehicle (UAV).



Tong Wei received the B.E. degree in remote sensing from the School of Remote Sensing and Information Engineering, Wuhan University, Wuhan, China, in 2023.

He is currently working toward the master's degree in photogrammetry and remote sensing with the School of Remote Sensing and Information Engineering, Wuhan University. His main research interests include computer vision and remote sensing images object detection and recognition.

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